Course Outline

1. Introduction
2. Neuronal Signaling
3. Visual System
4. Computational Methods I
5. Decision making
6. Multi-modal Sensory Systems
7. Hippocampus, Spatial Cognition & Memory
8. Working memory
9. Motor Systems
10. Computational Methods II
11. Learning & Plasticity
12. Value and Reward
13. Project Presentation
14. Final Q&A

系统和计算神经科学导论

六、多模态感觉系统

Multi-Modal Sensory Systems

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- Auditory
- Vestibular
- Somatosensory
- Multimodal Integration
- Causal Inference

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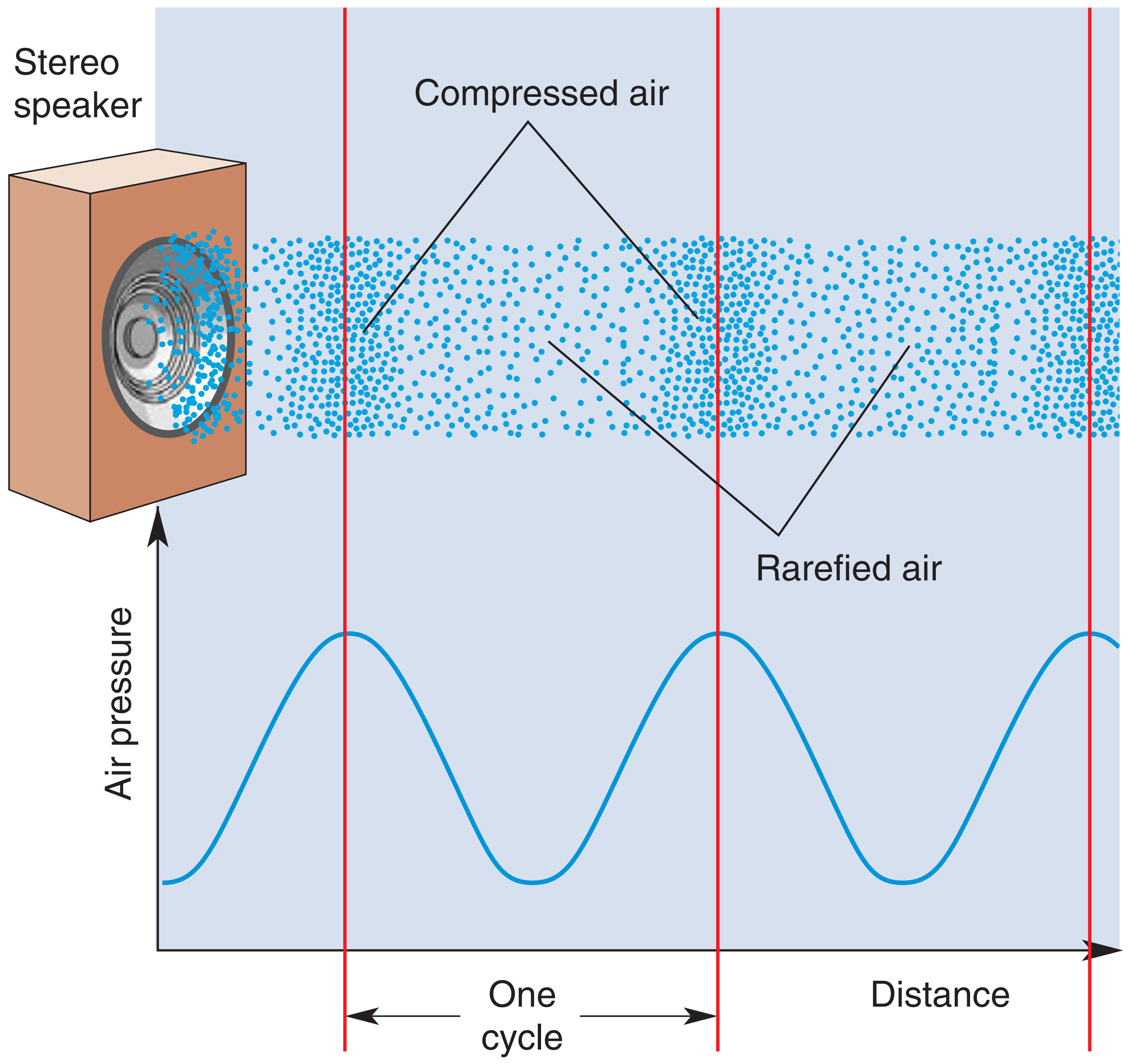
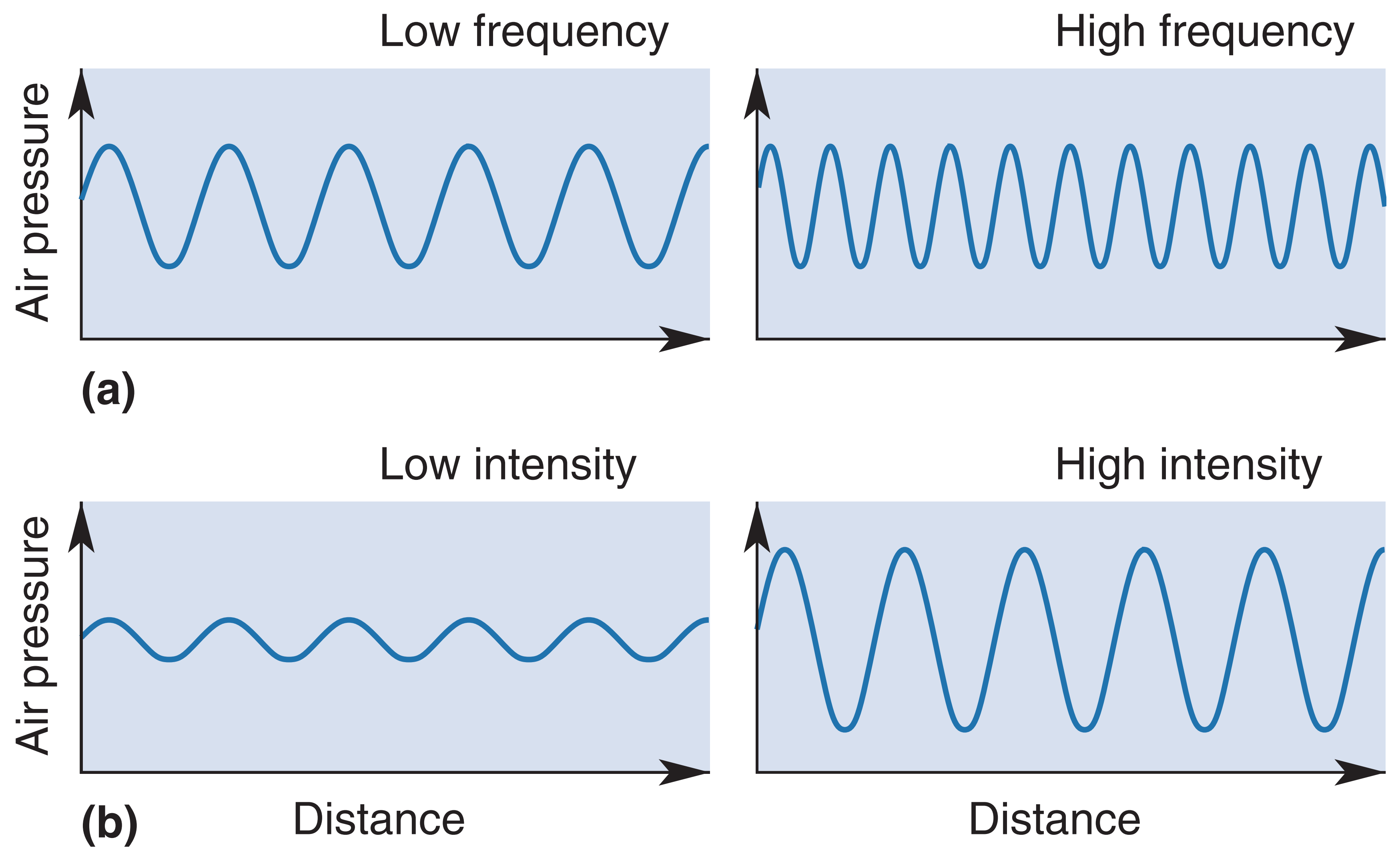
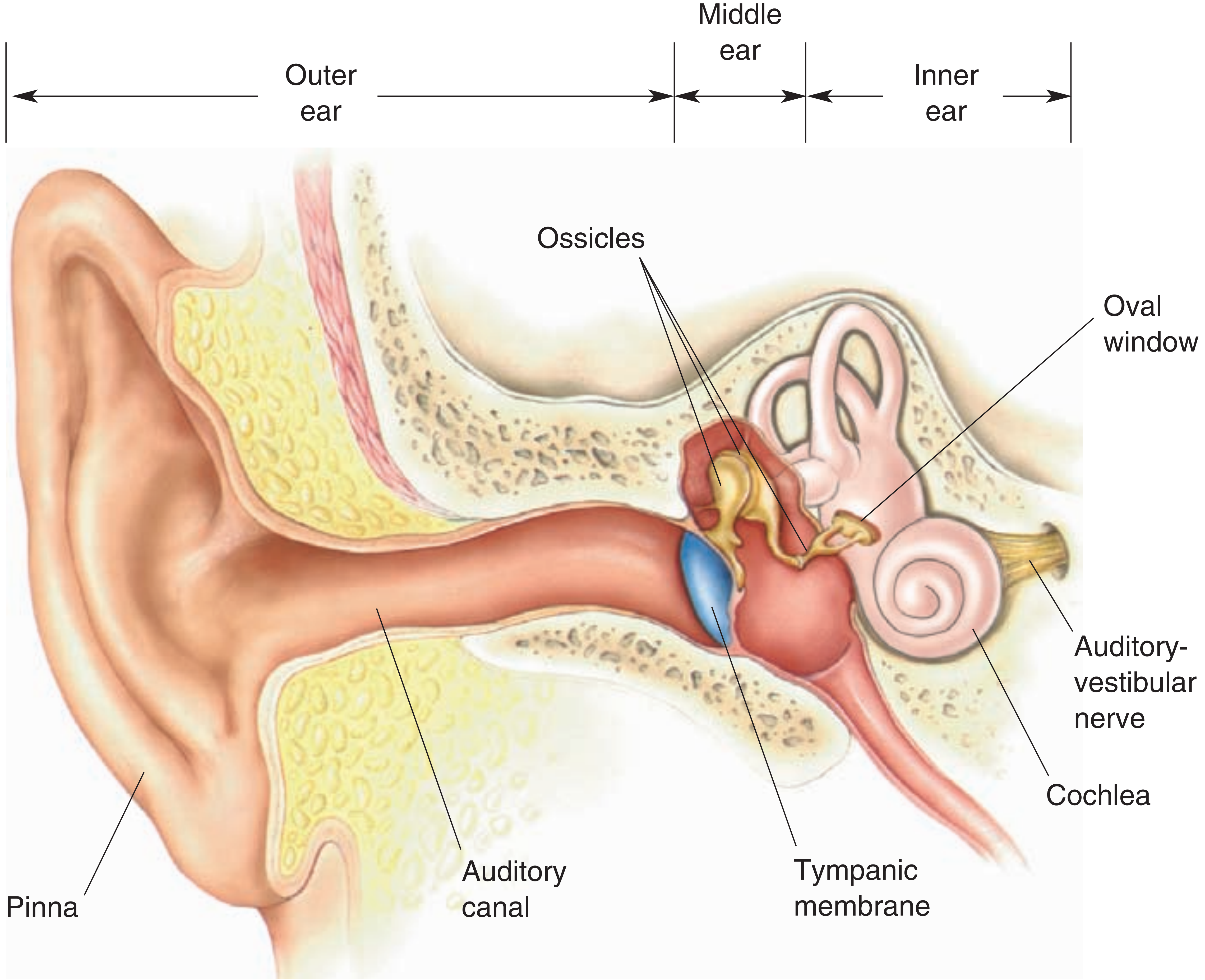
Sound Waves

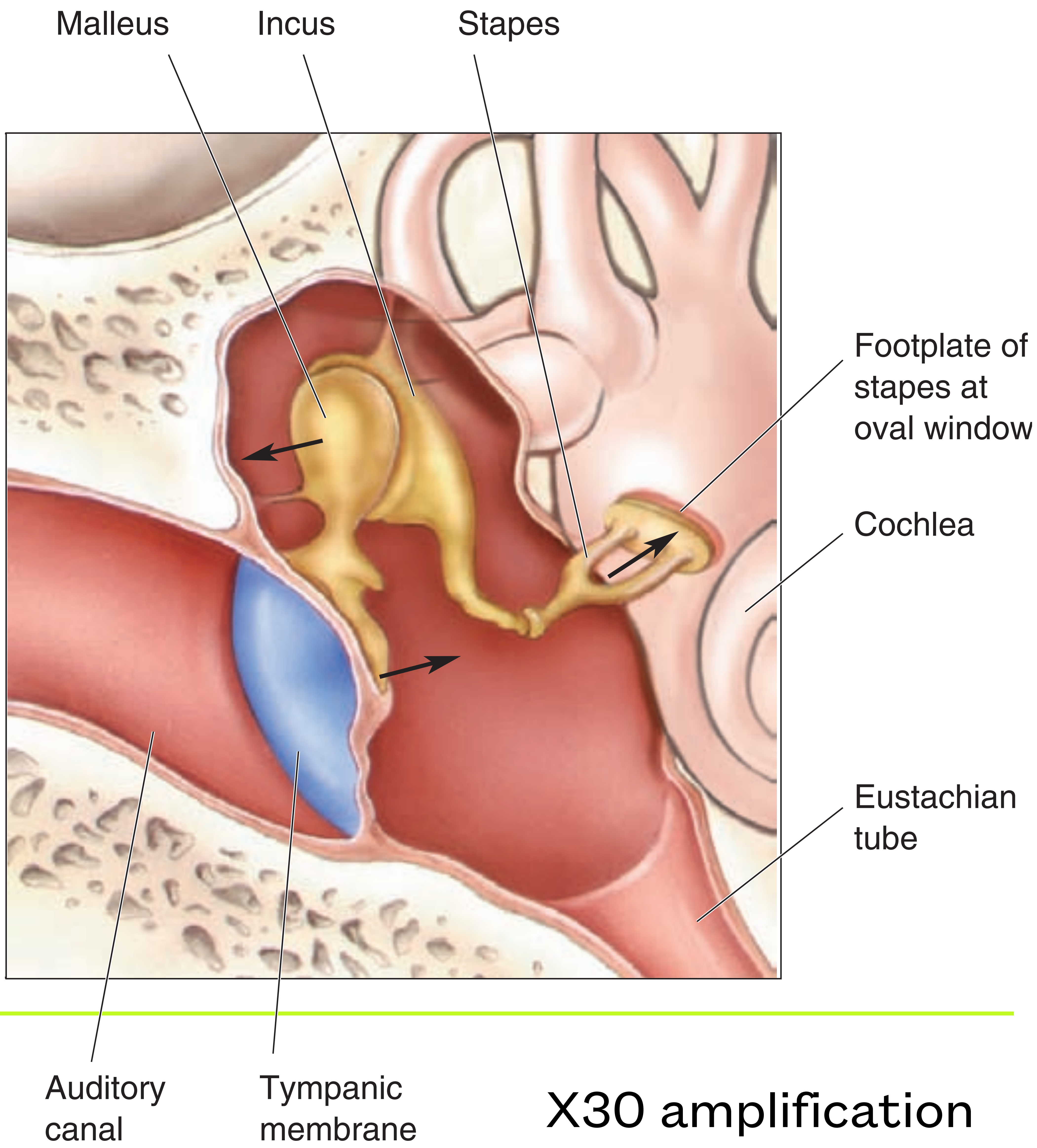




20-20k Hz

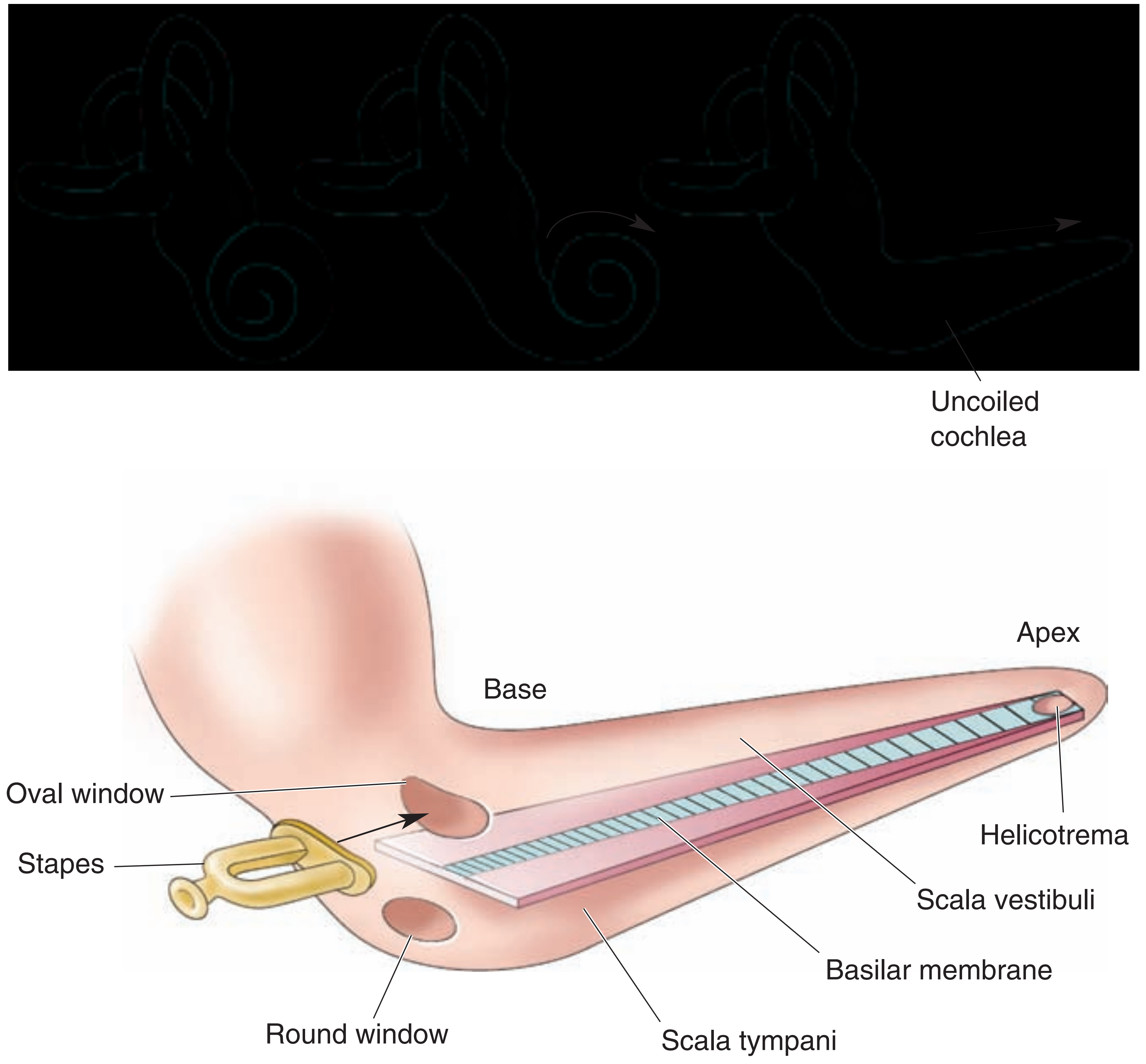
Outer-Middle-Inner Ear



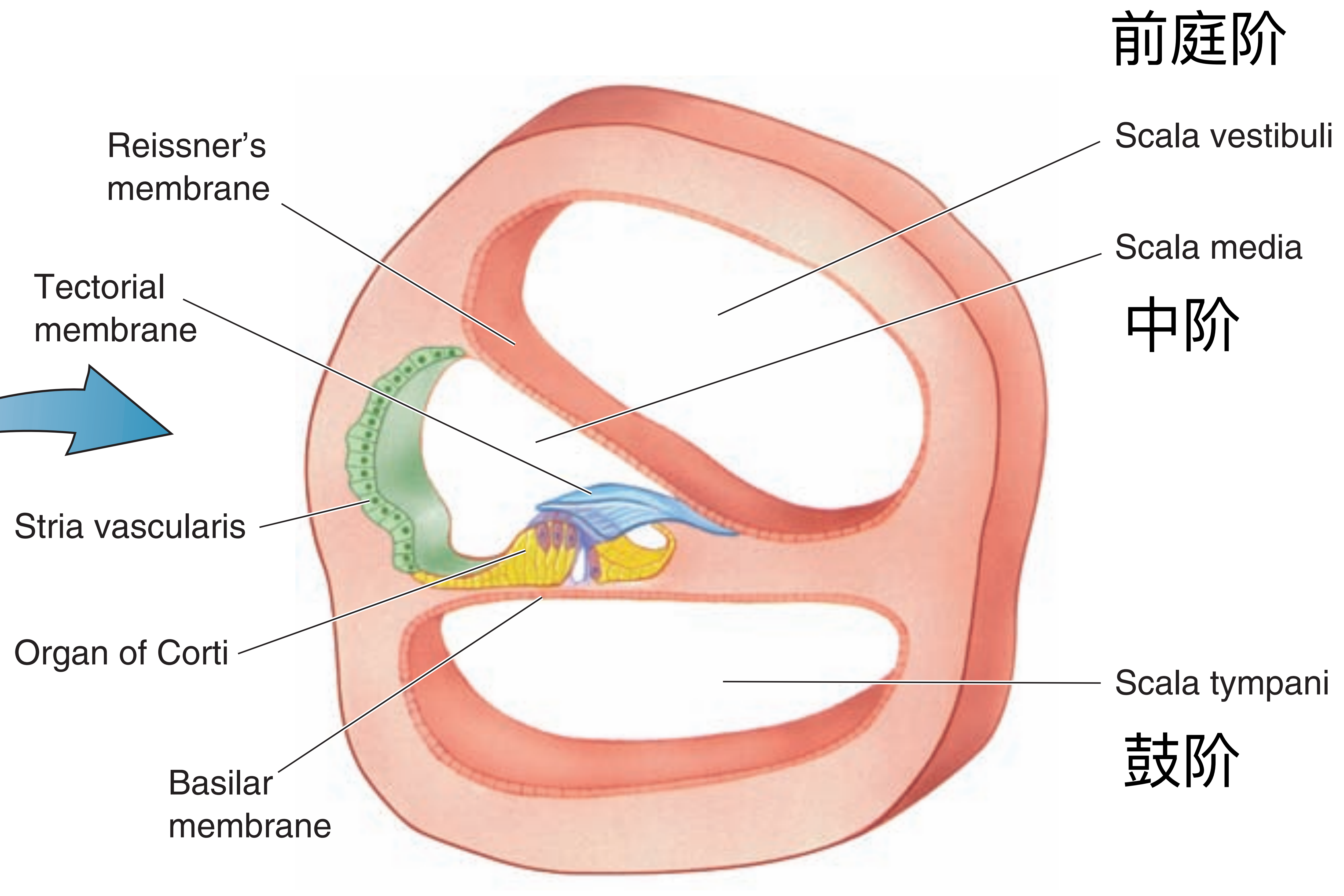


X30 amplification

Cochlea

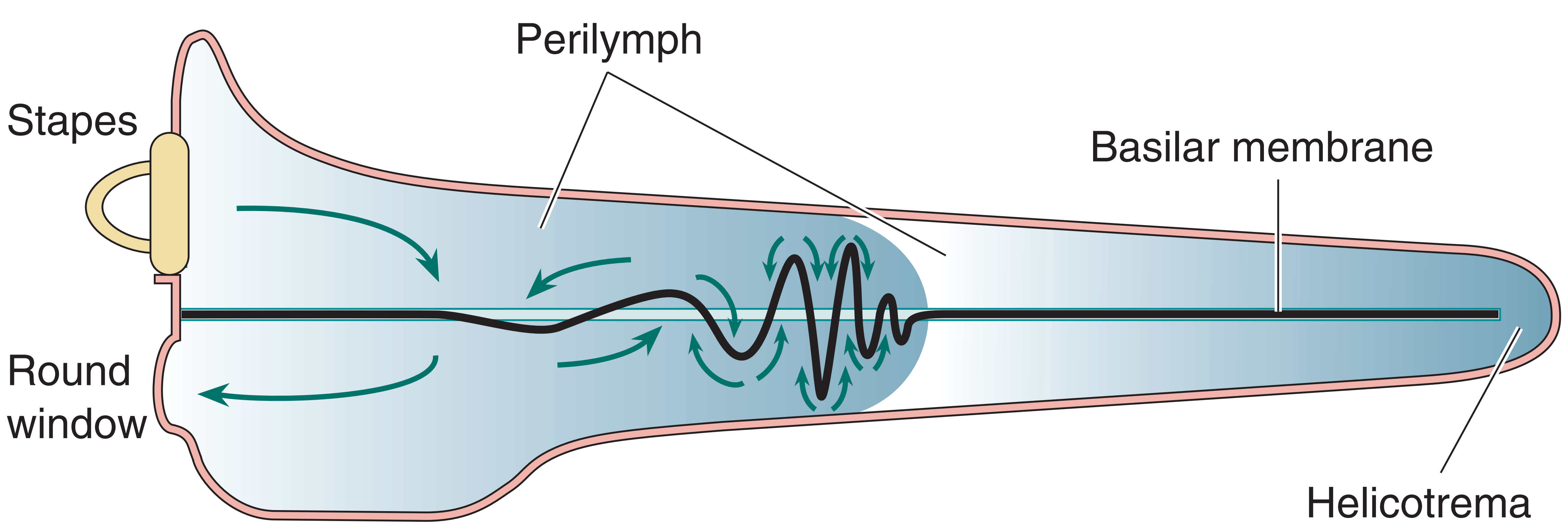


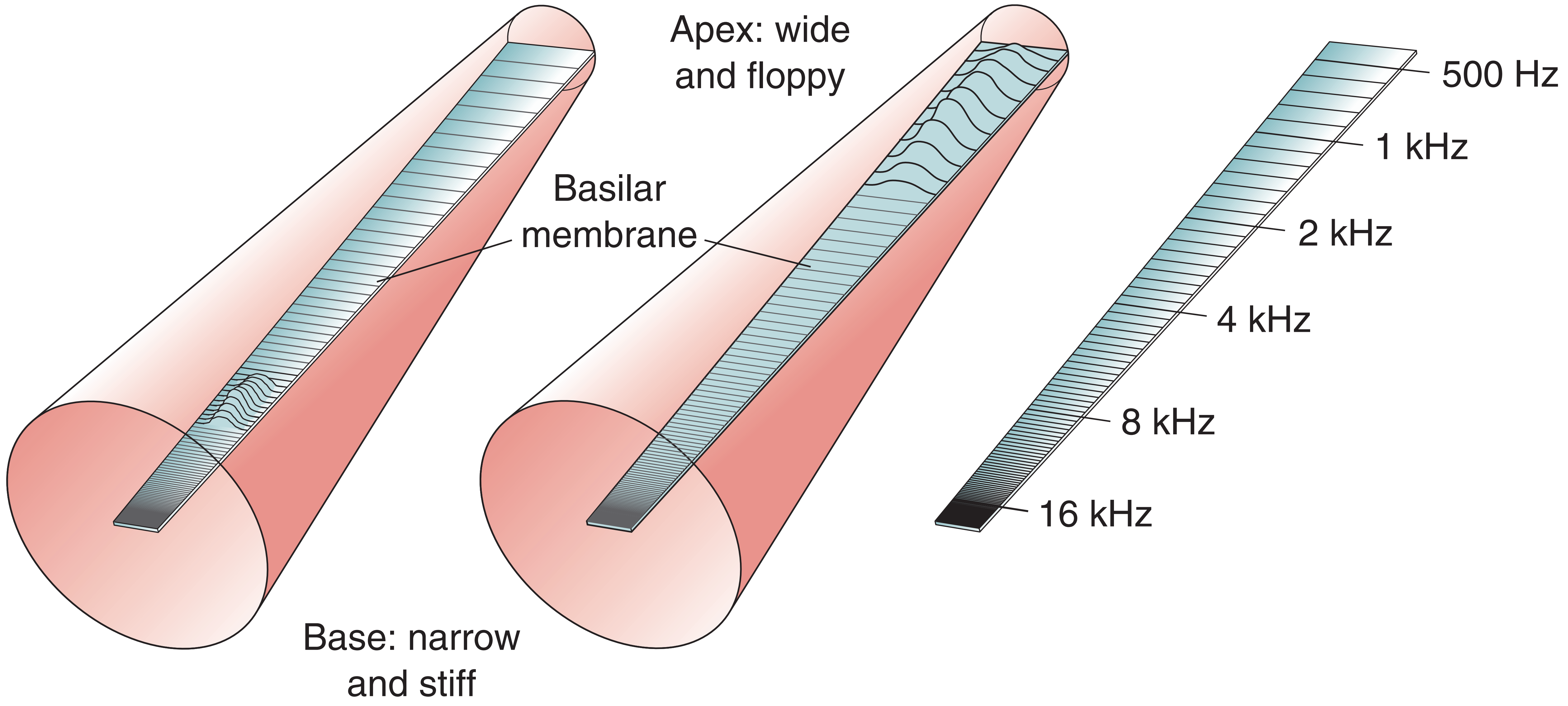
Uncoiled
cochlea
Apex
Base
Oval window
Stapes
Helicotrema
Scala vestibuli
Basilar membrane
Round window
Scala tympani



前庭阶
Reissner's
membrane
Scala vestibuli
Scala media
中阶
Tectorial
membrane
Stria vascularis
Organ of Corti
Scala tympani
鼓阶
Basilar
membrane

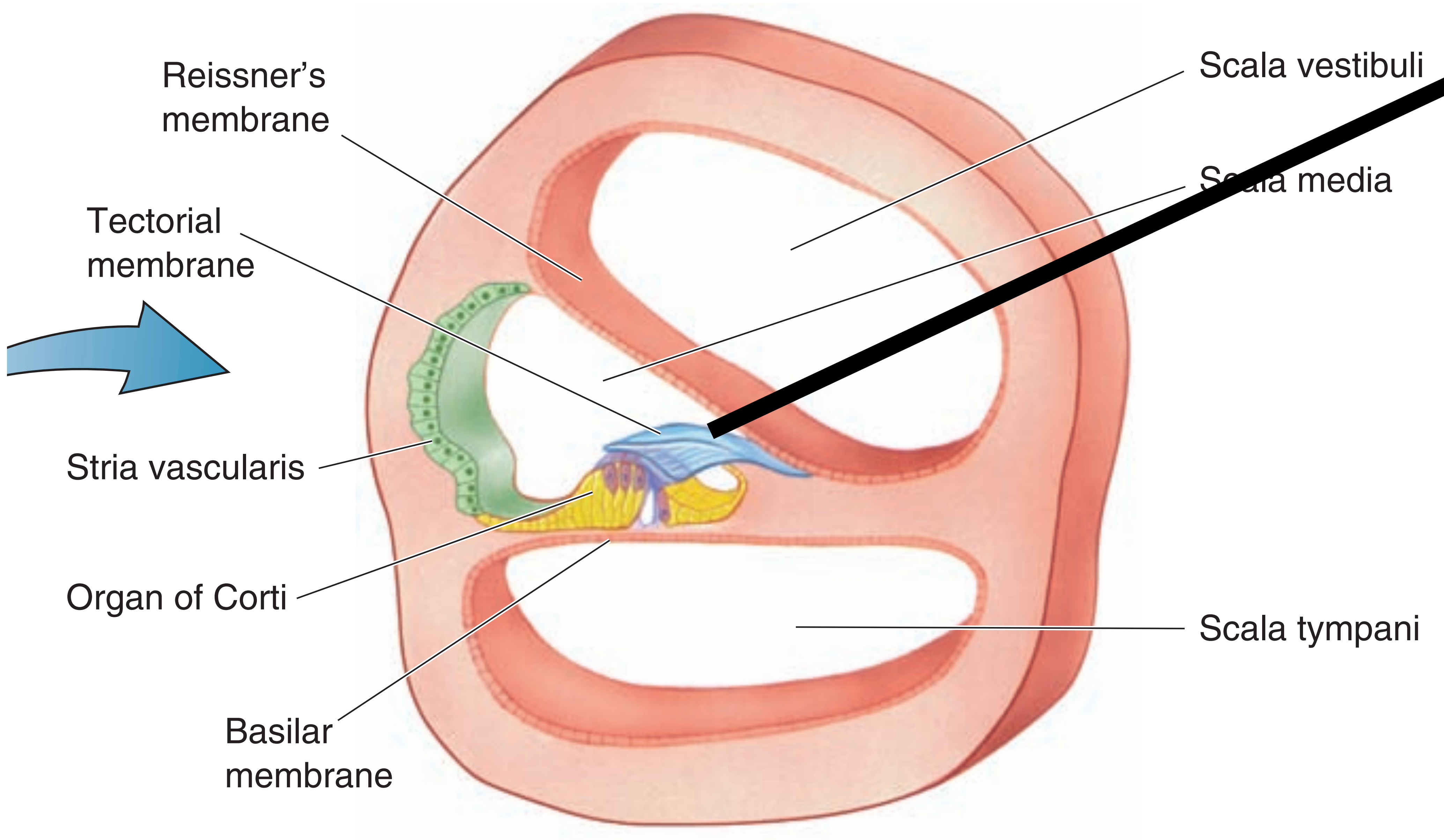
Cochlea



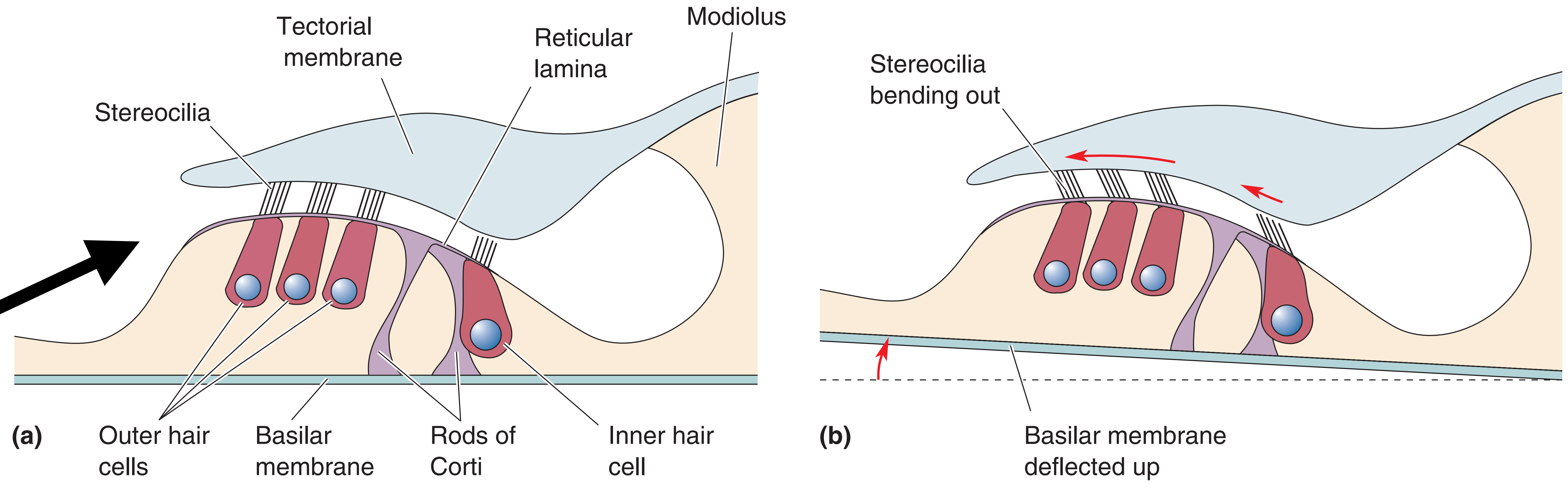


(a) High frequency **(b)** Low frequency **(c)** Frequency producing maximum amplitude

Organ of Corti



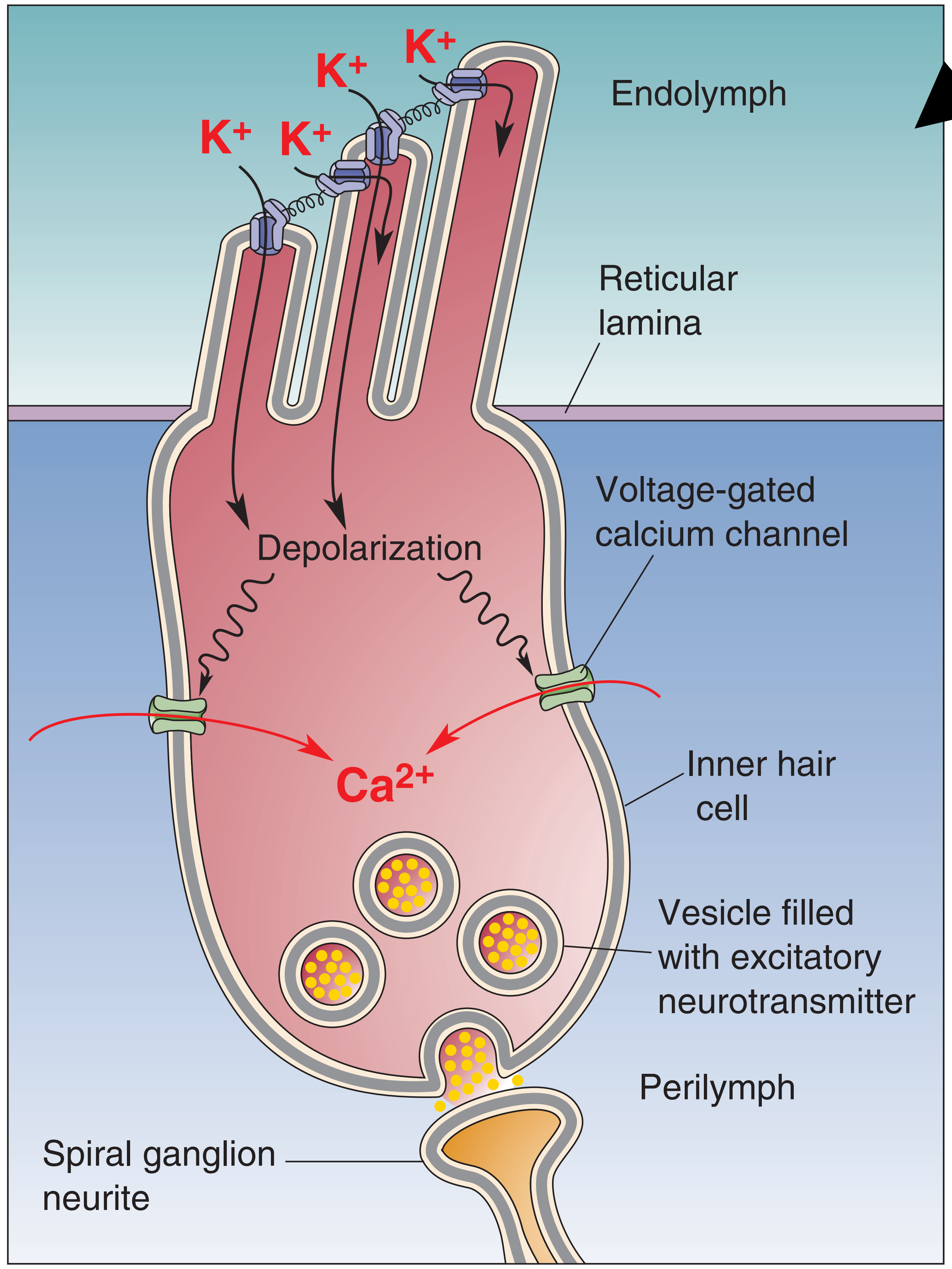
科尔蒂器

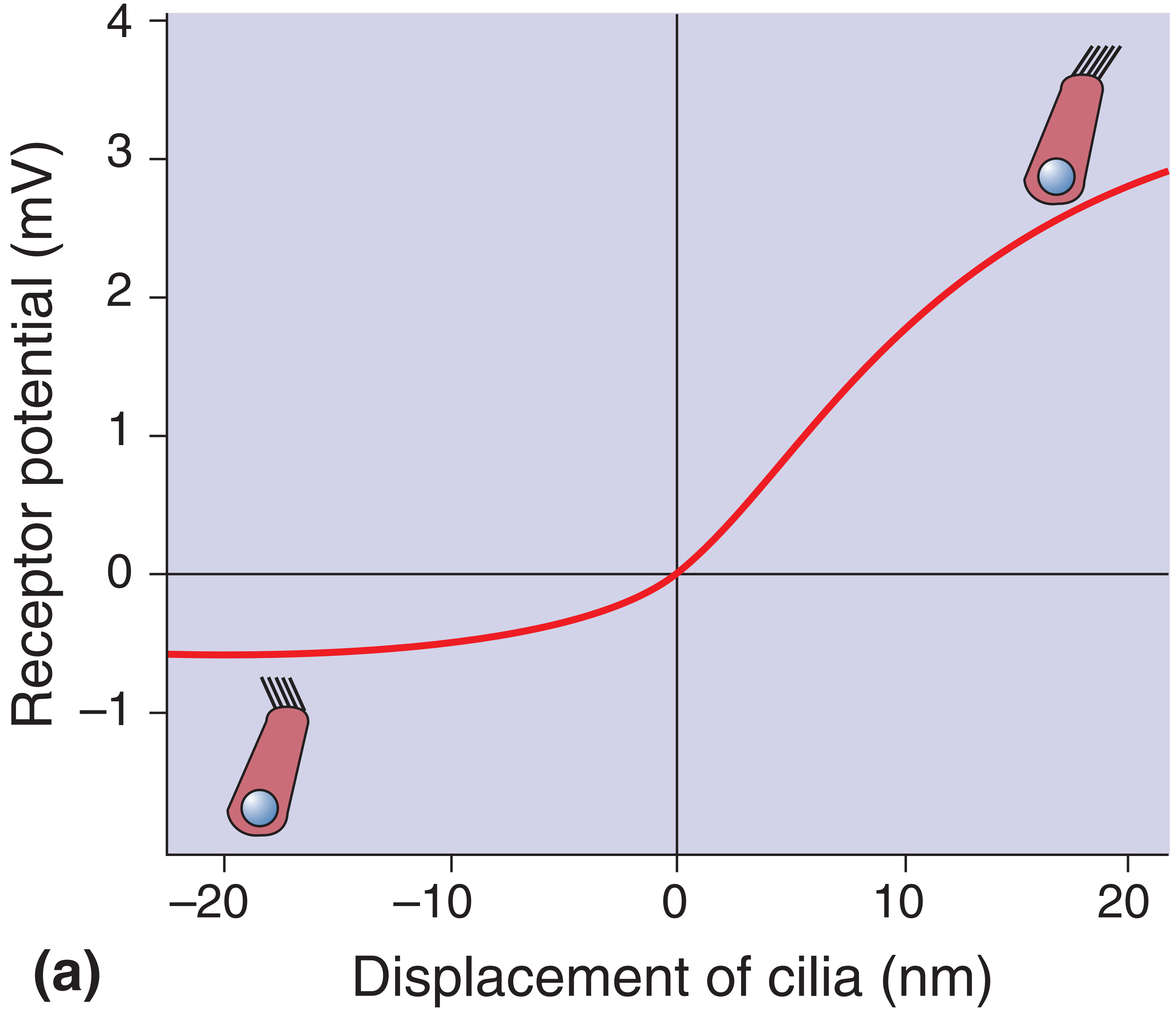


- **Outer hair cells** fine-tune and amplify the mechanical vibrations, increasing the sensitivity and selectivity of the cochlea to sound frequencies.
- **Inner hair cells** transduce these amplified vibrations into electrical signals

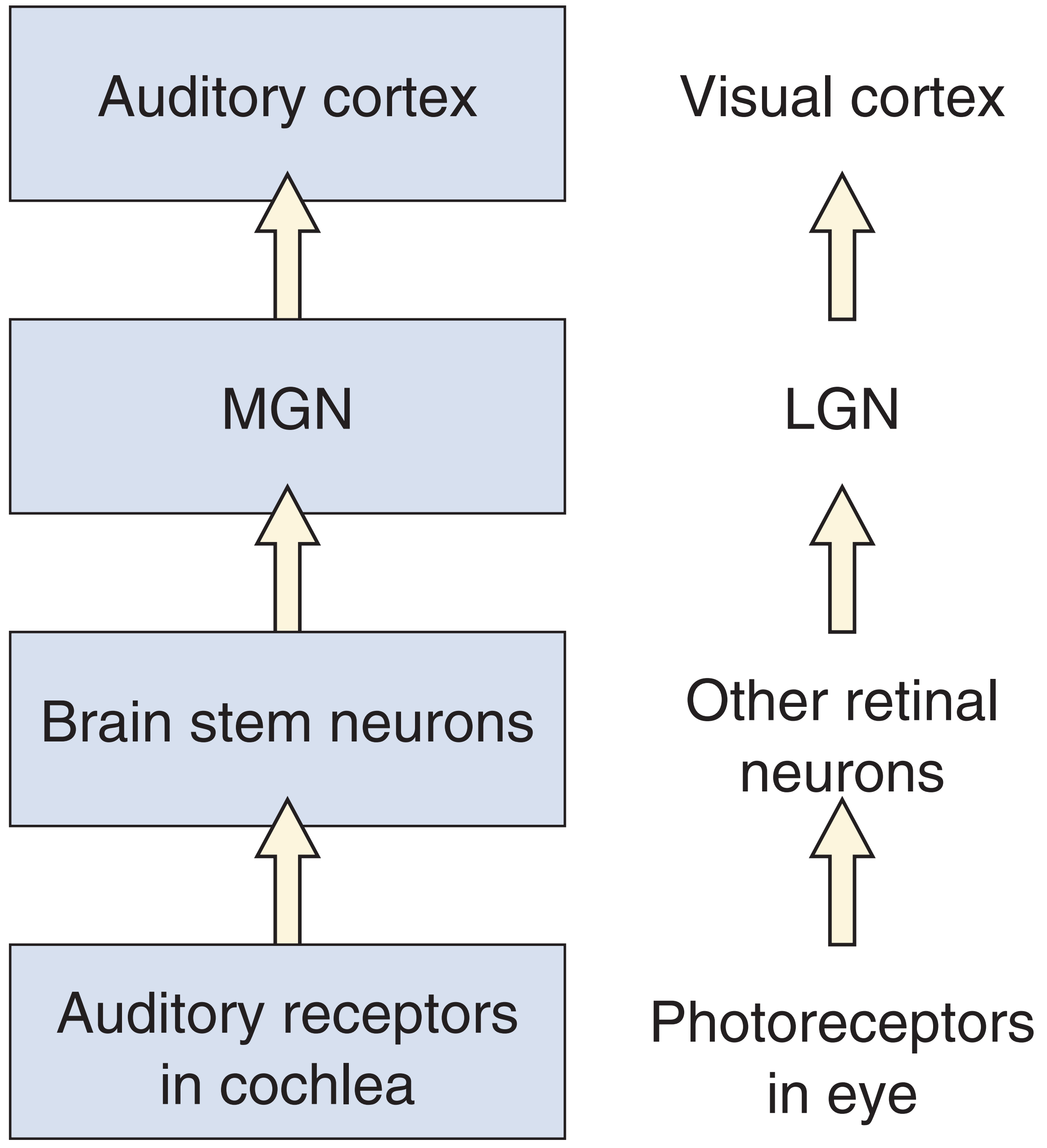
Hair Cells

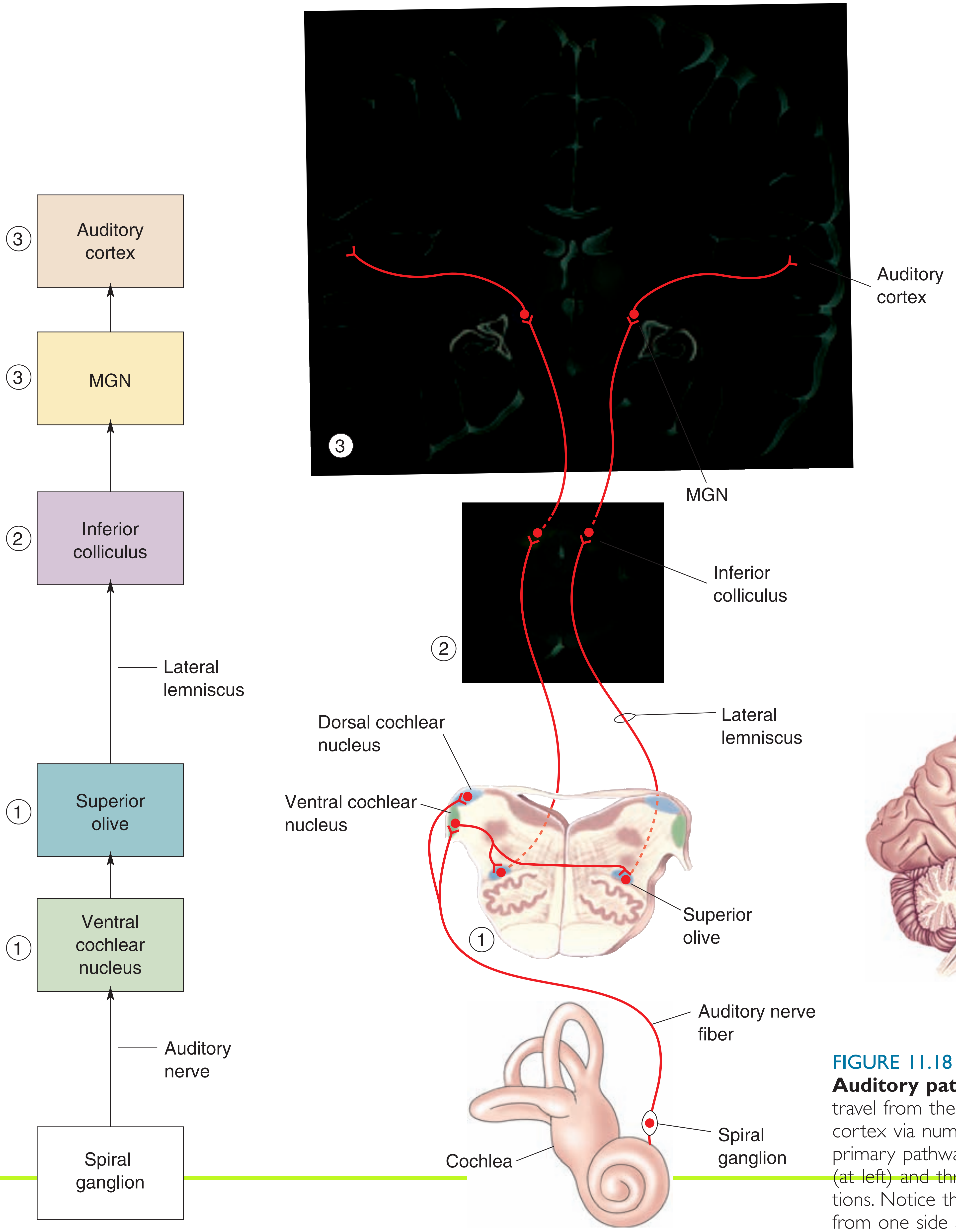
高浓度K+





Auditory Pathway





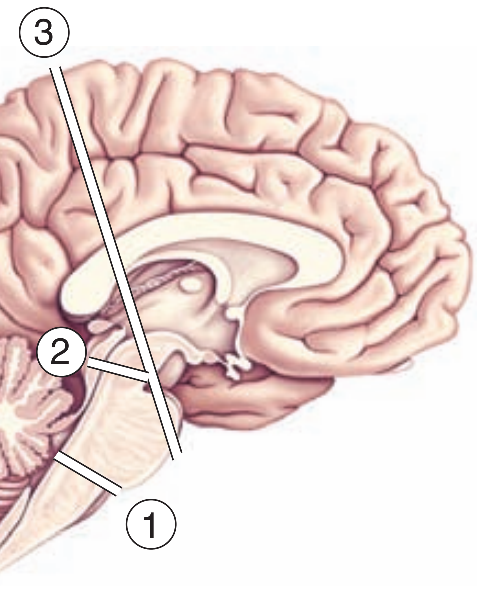
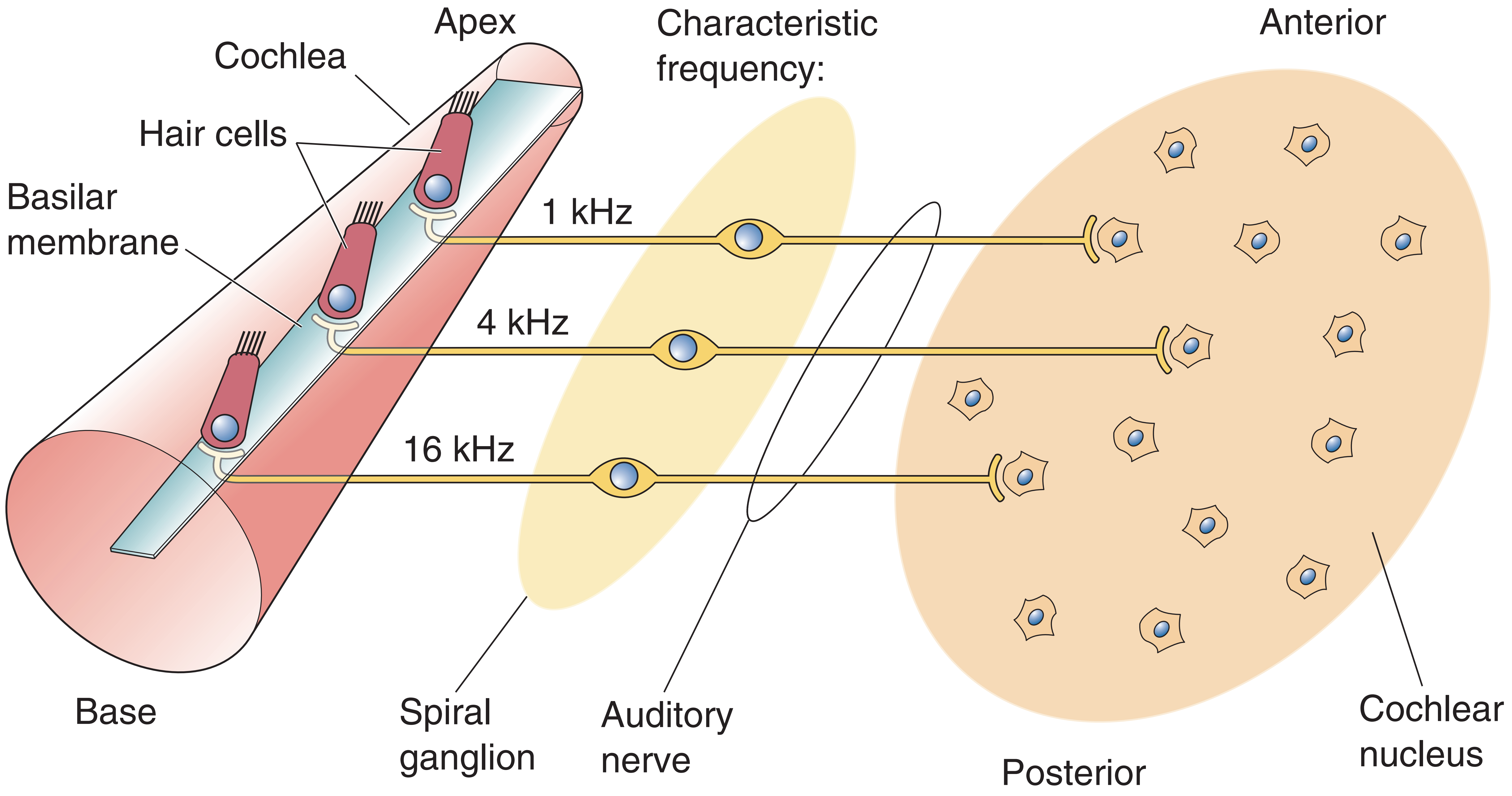
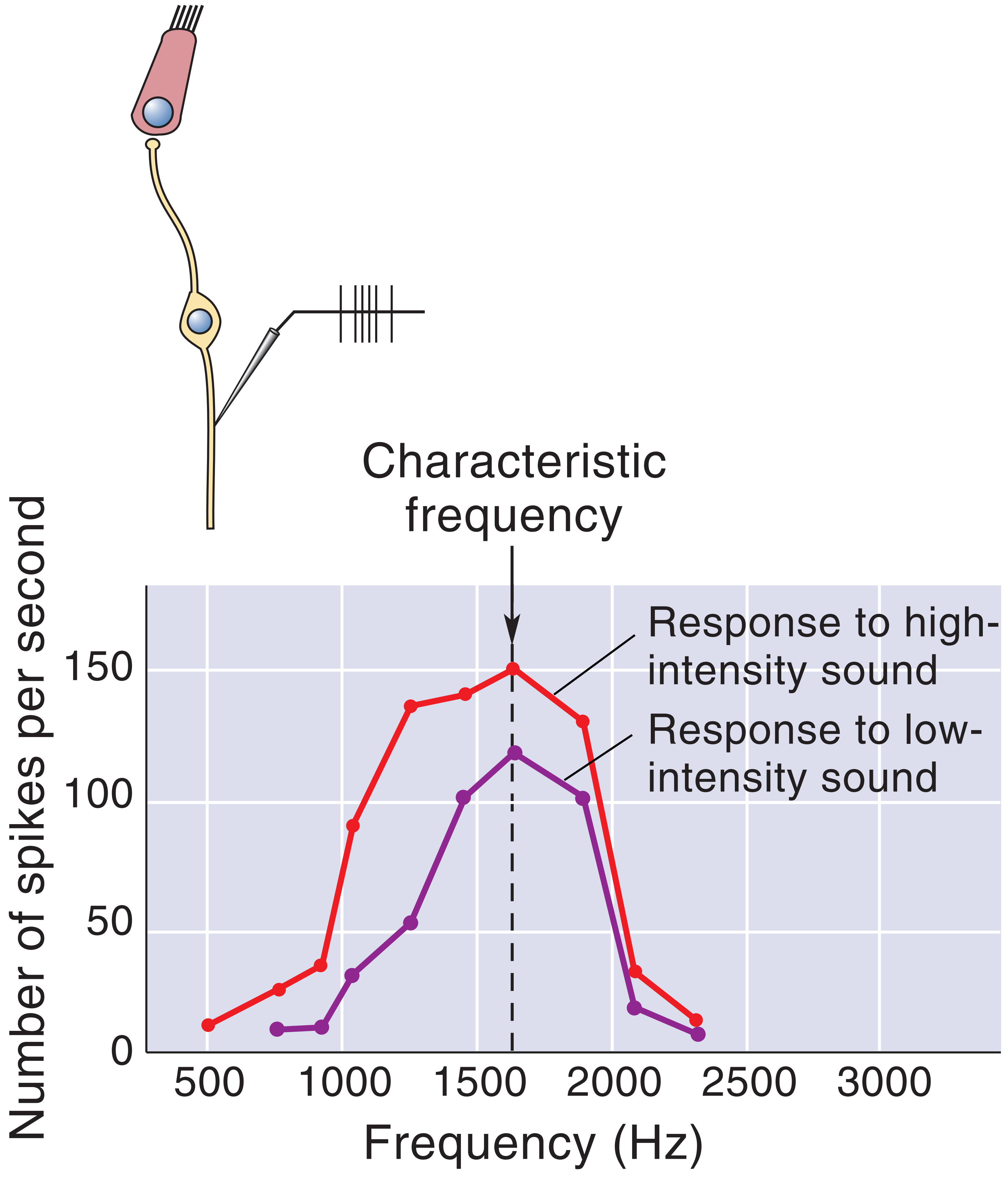


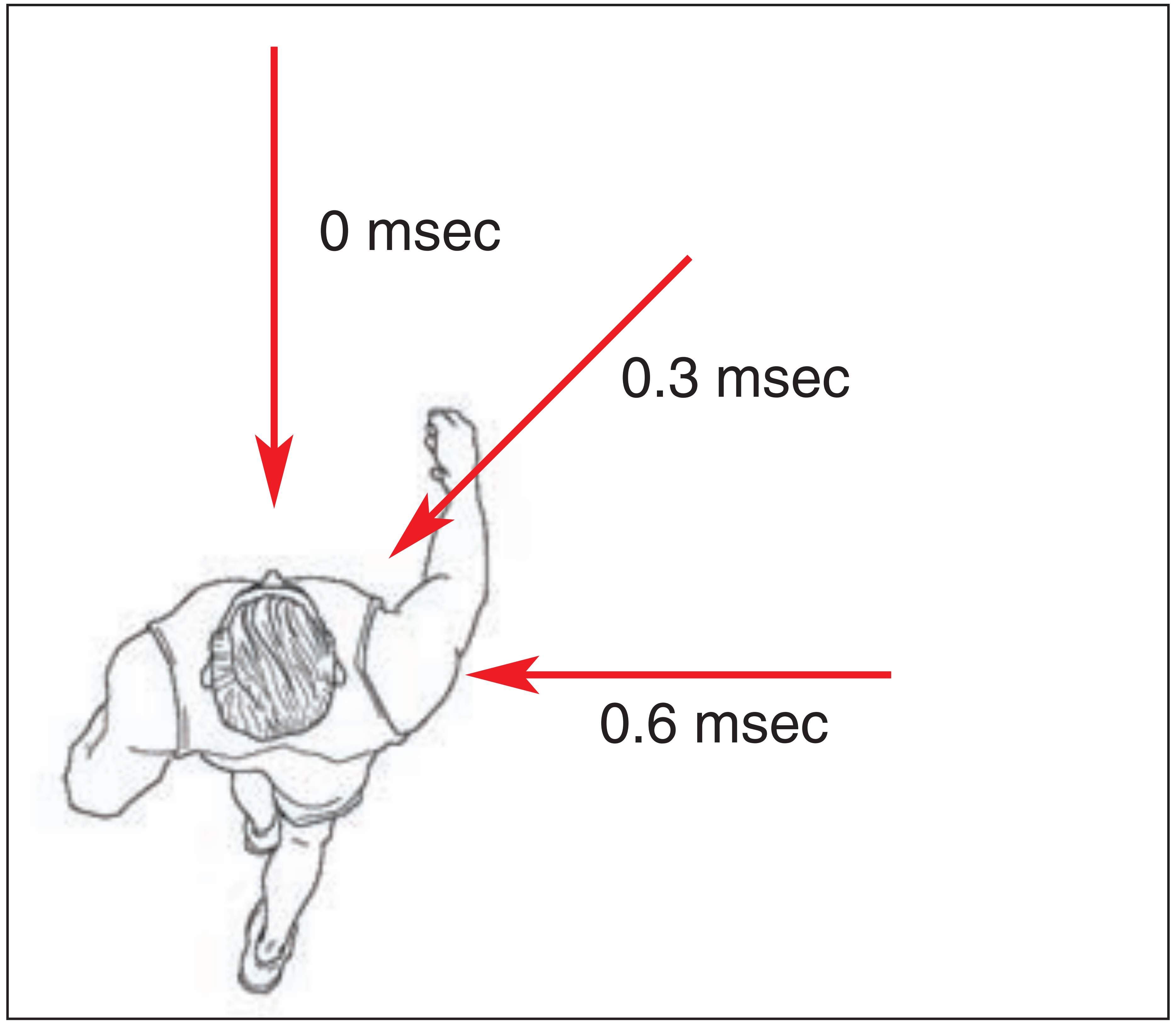
FIGURE 11.18
Auditory pathways. Neural signals can travel from the spiral ganglion to auditory cortex via numerous pathways. Here, a primary pathway is shown schematically (at left) and through brain stem cross sections. Notice that only the connections from one side are illustrated.

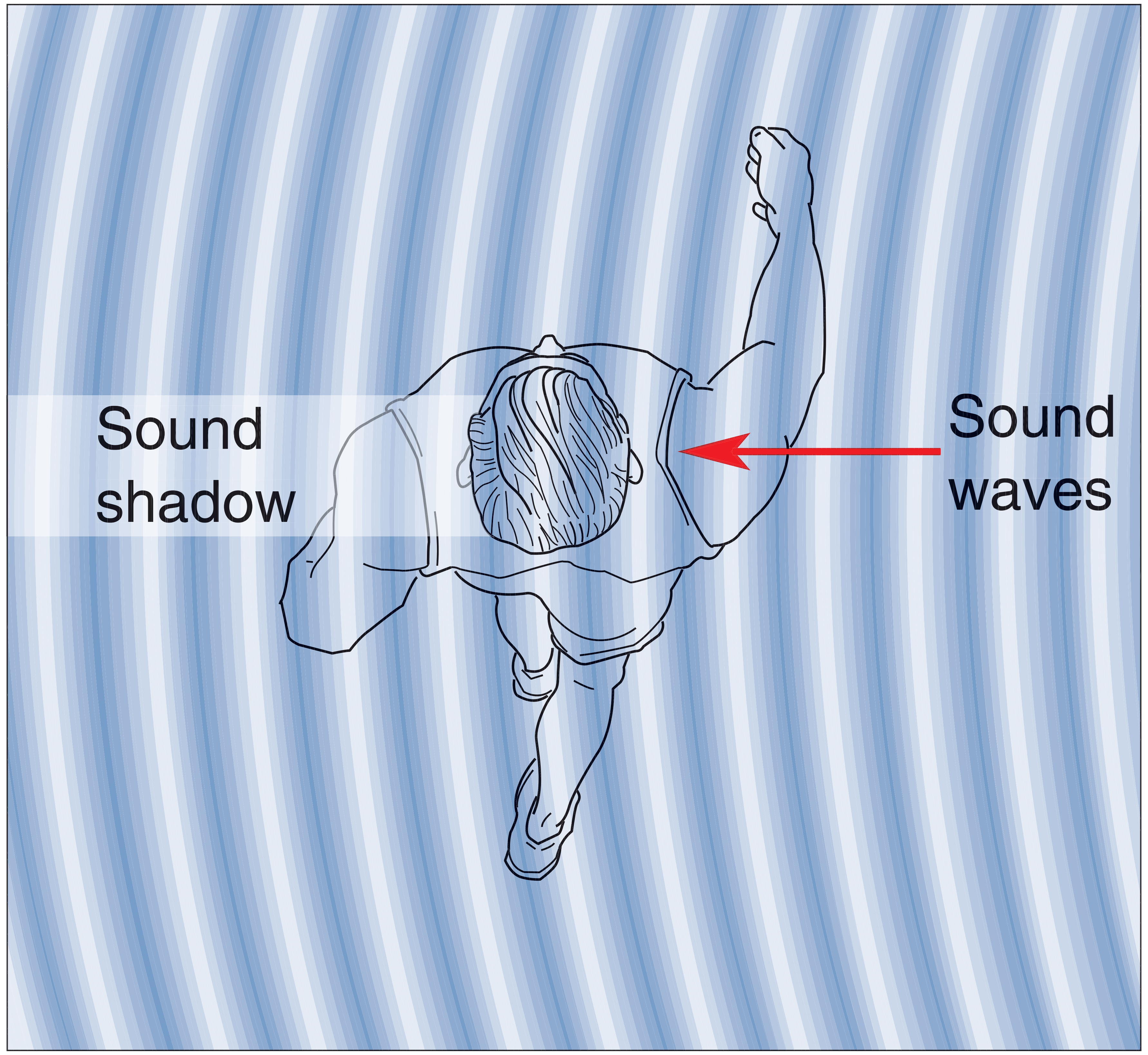
Auditory Pathway





Sound localization

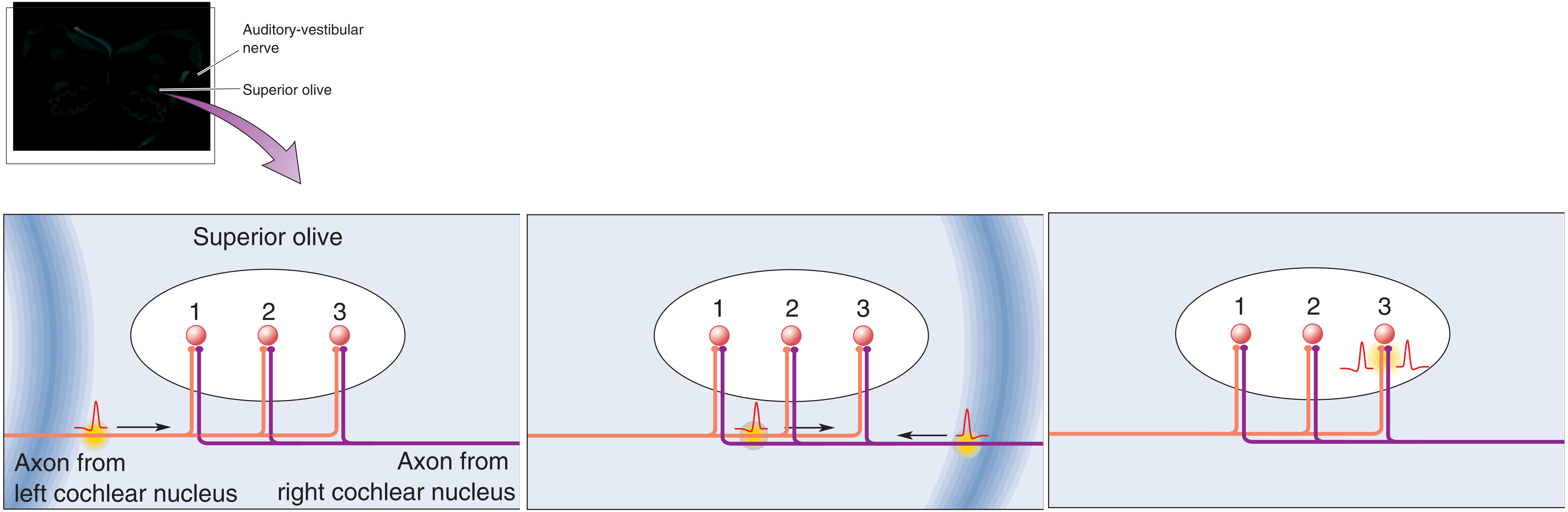




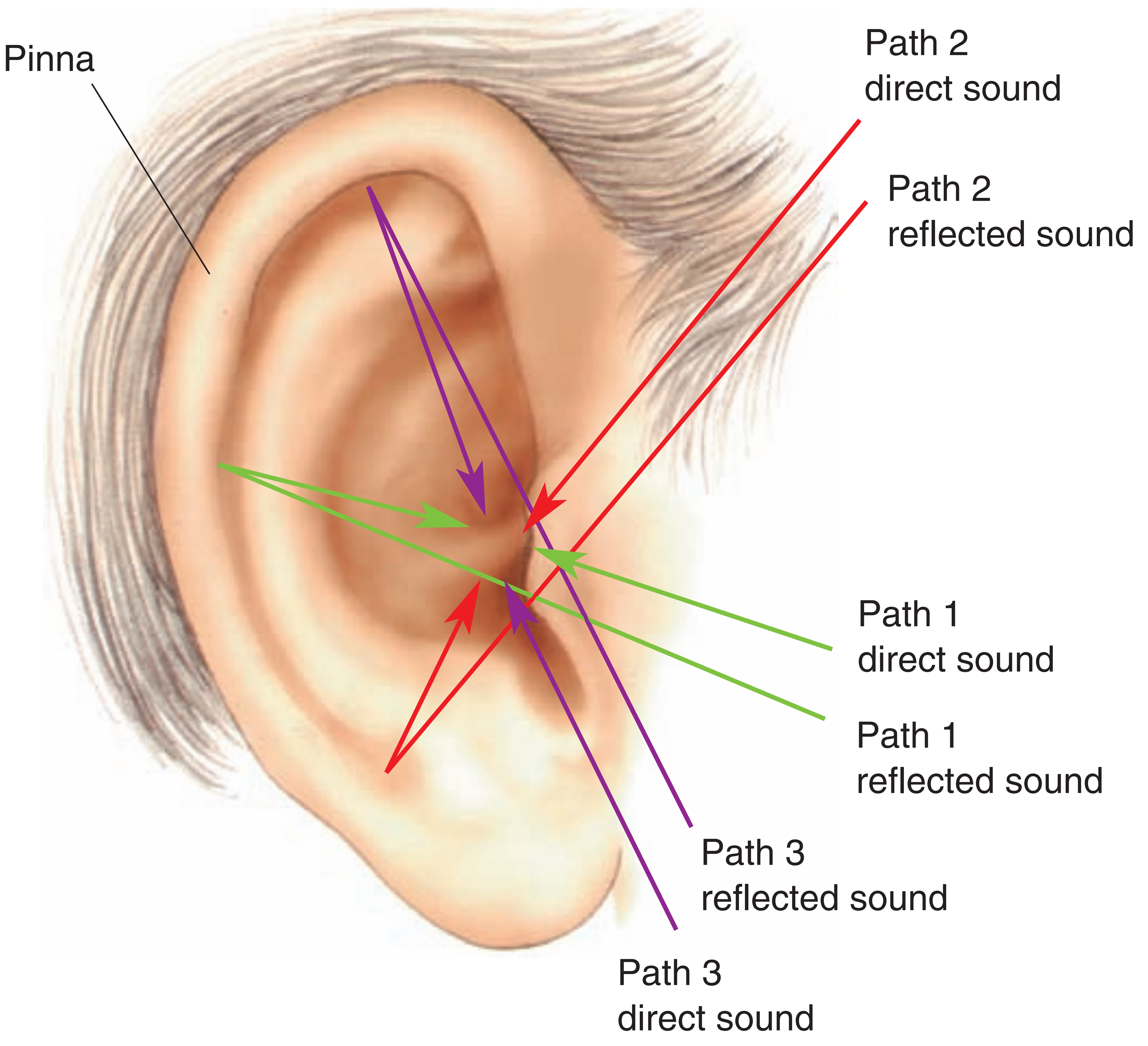
Interaural Time Delay
20~2000 Hz

Interaural Intensity Difference
2000-20000 Hz

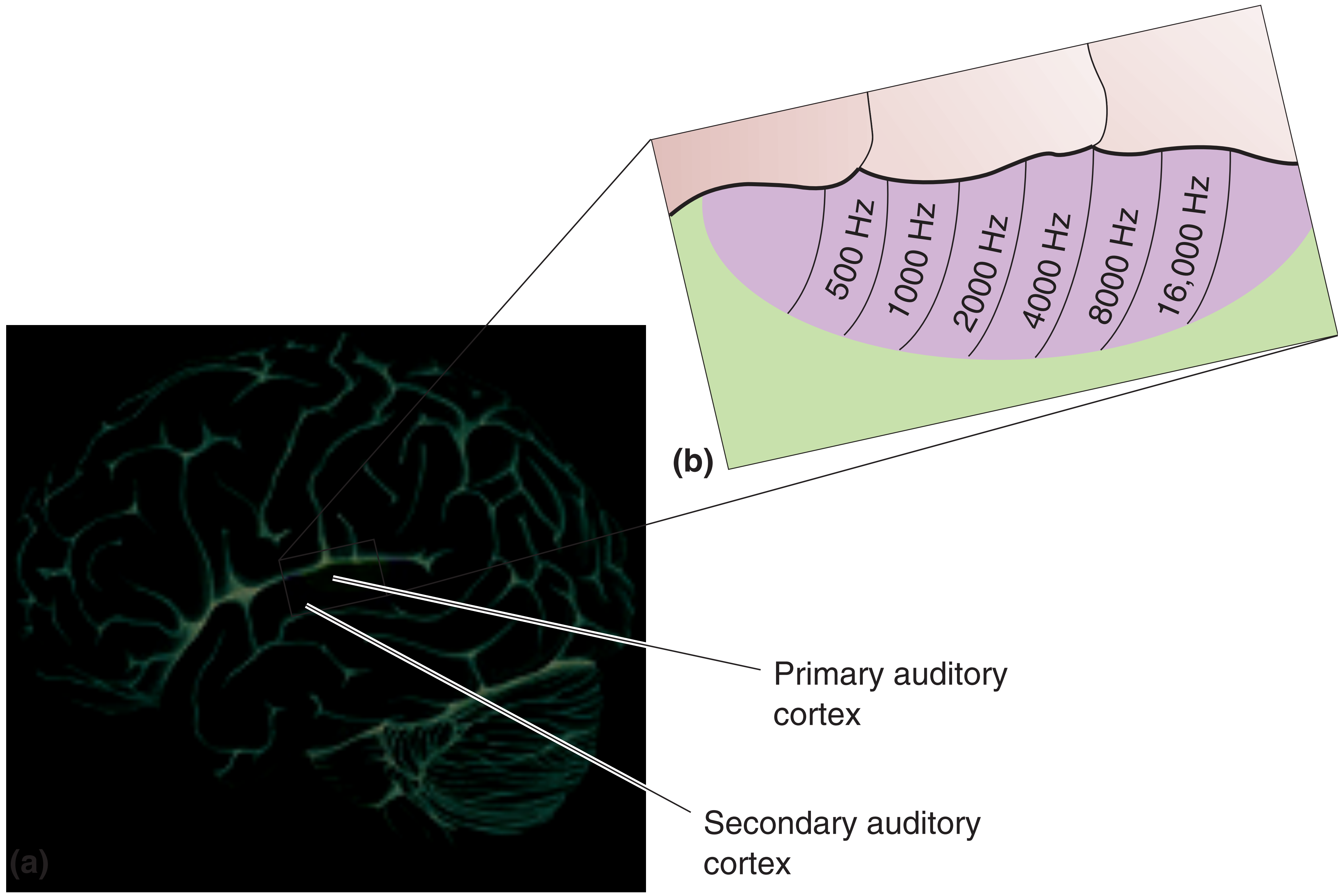
Sound localization



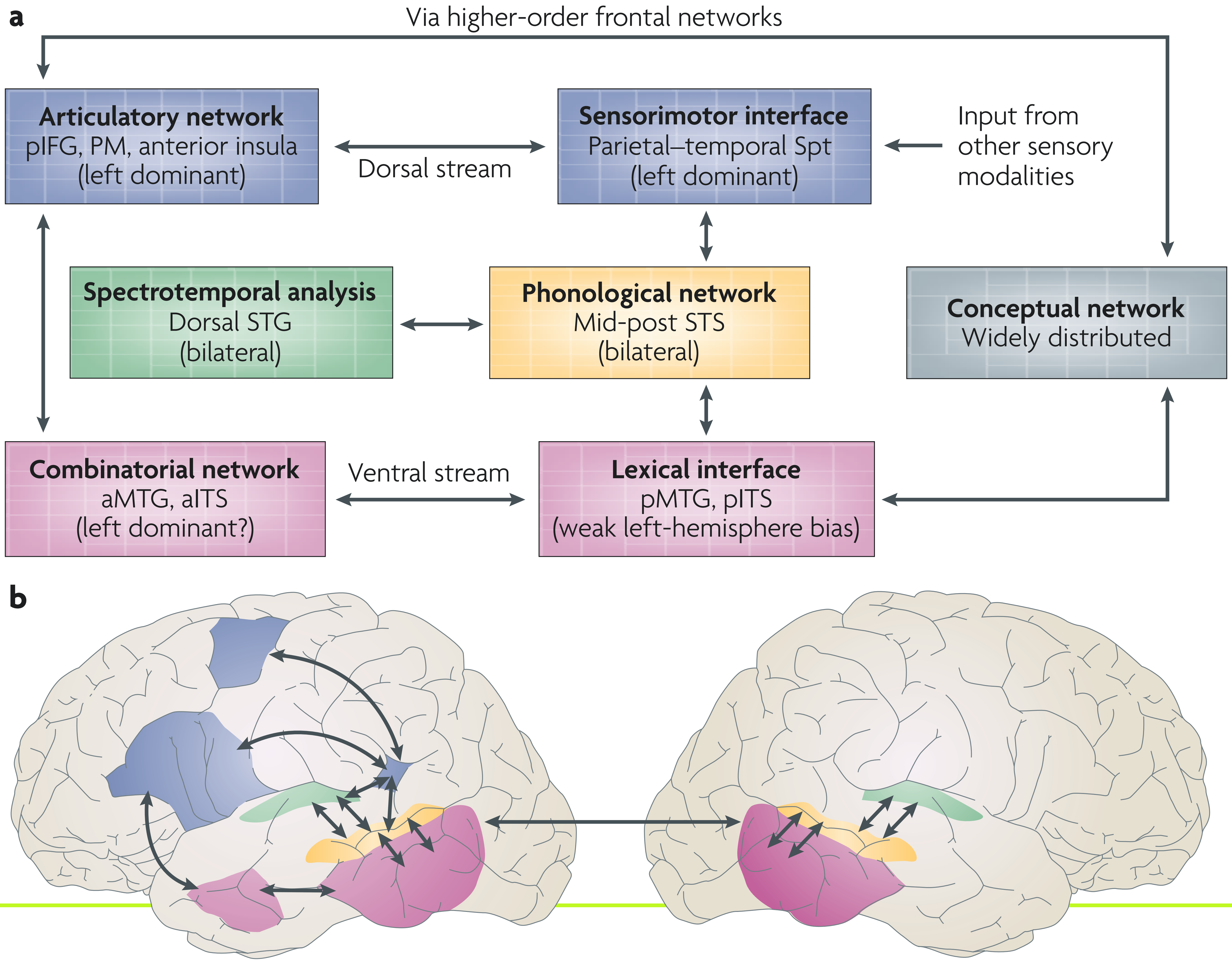
Sound localization: Vertical



Auditory Cortex



Language Circuitry



Hickok & Poeppel, 2007

- Auditory
- **Vestibular**
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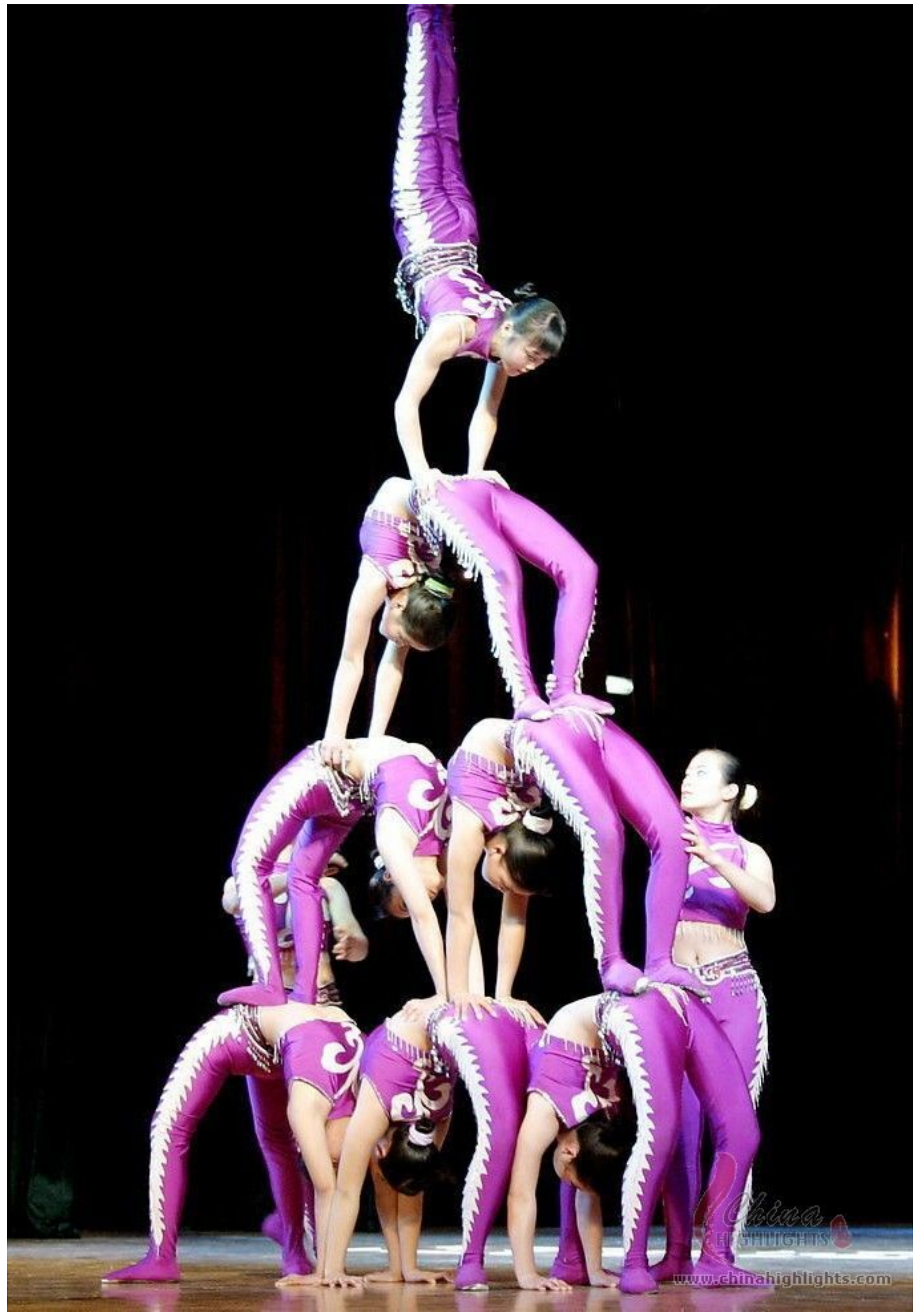
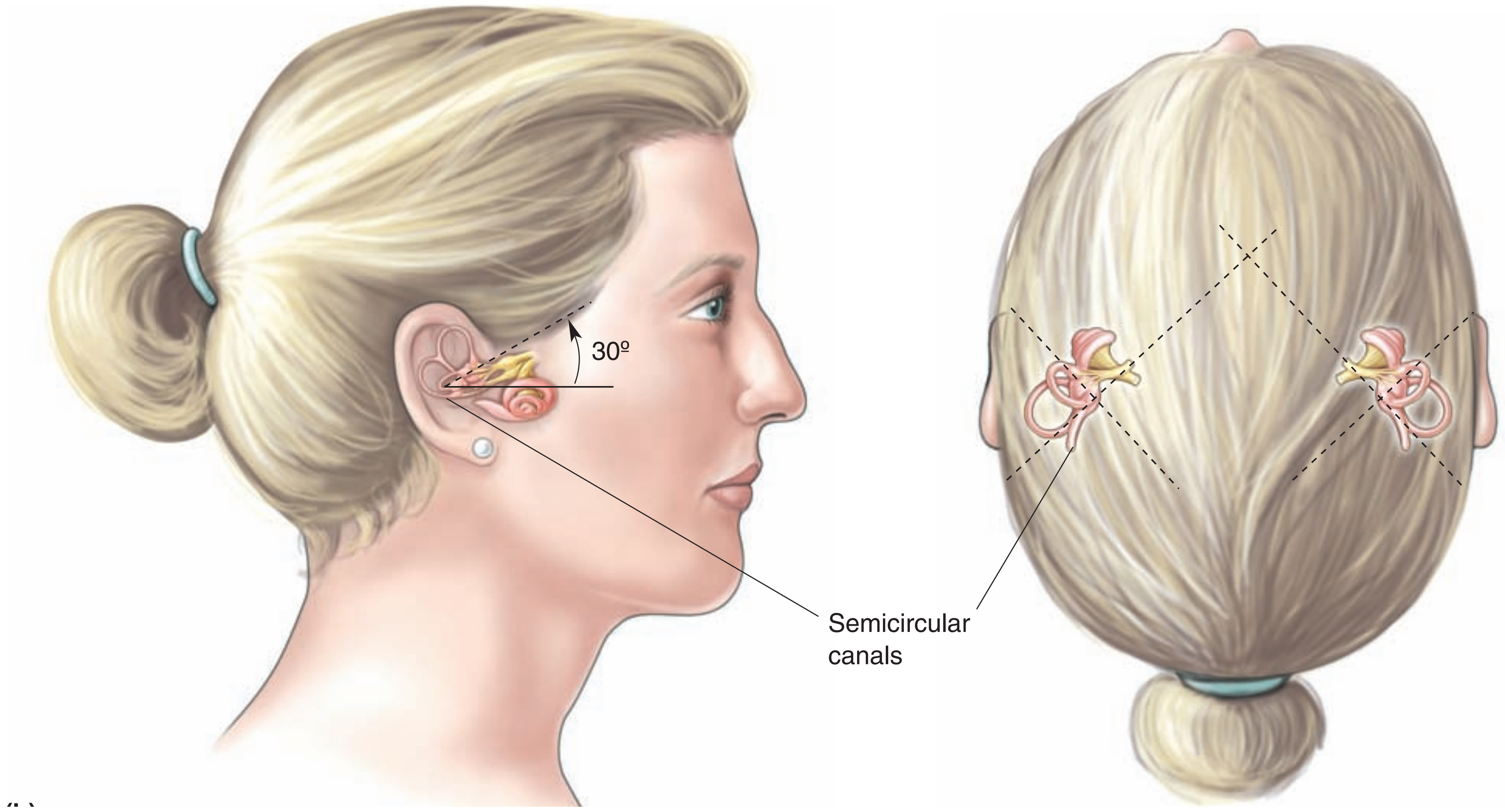
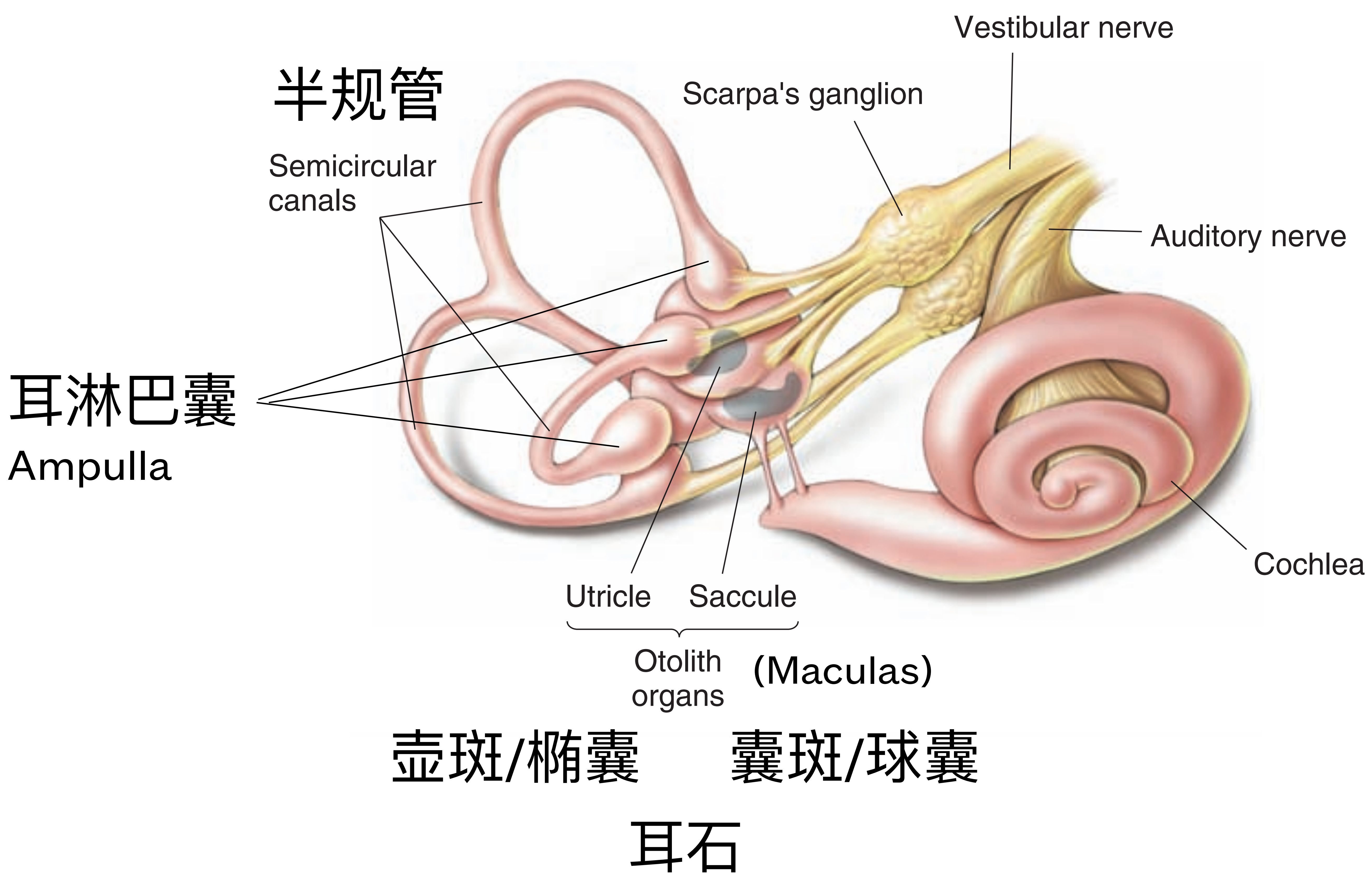
Vestibular System 前庭系统

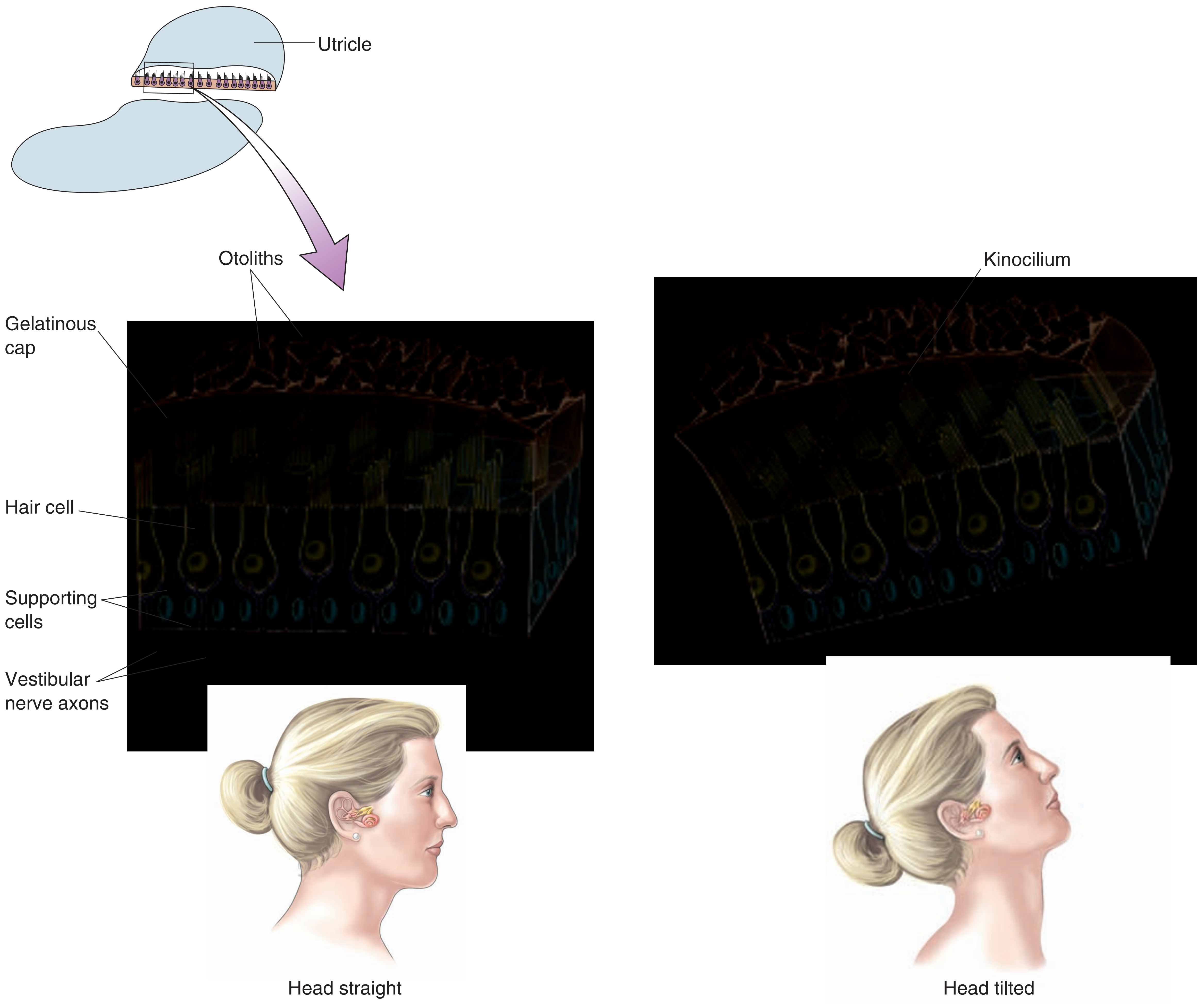




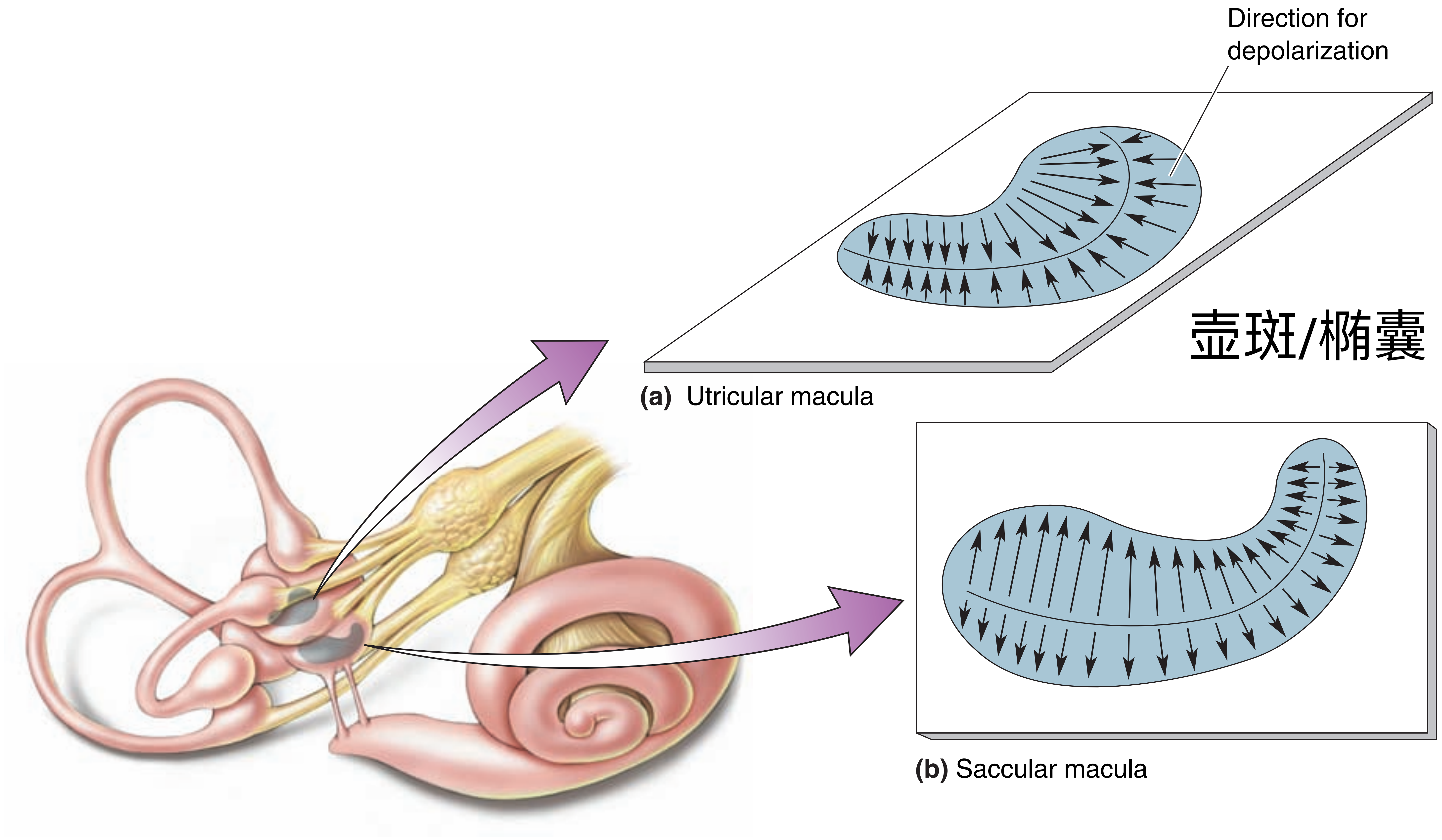
Vestibular Organ



Macular Hair Cells



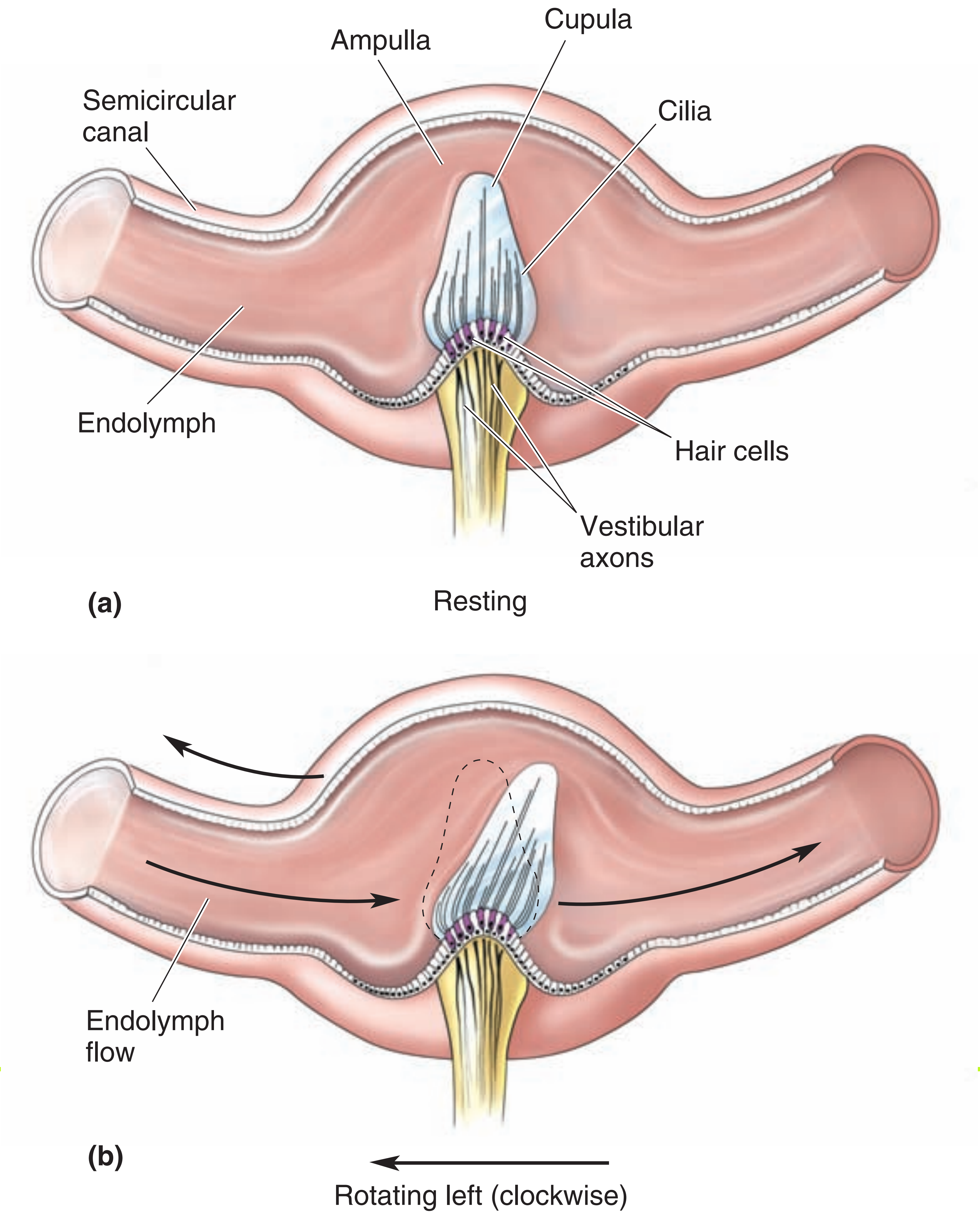
Macular Hair Cells

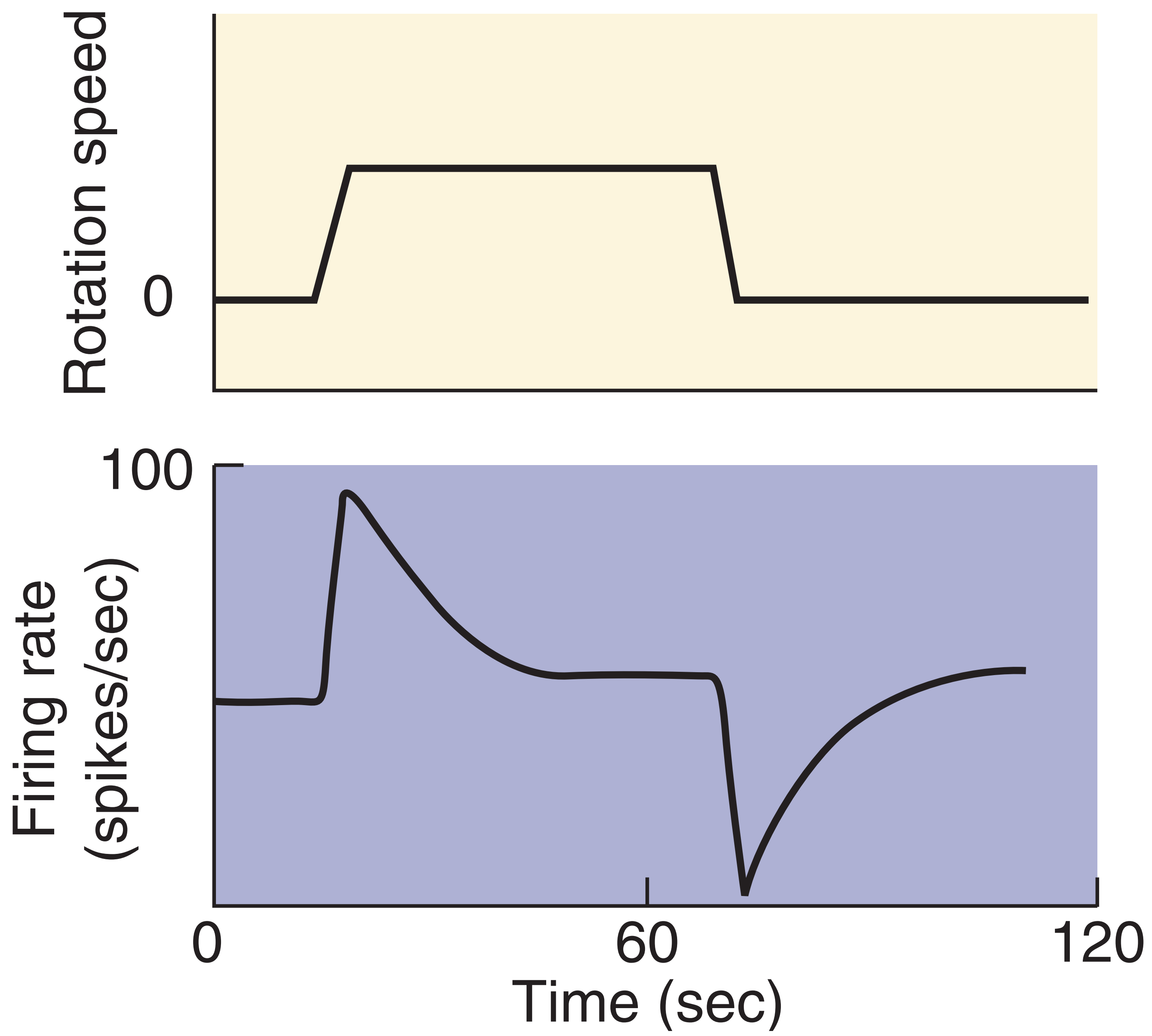


囊斑/球囊

感受头倾斜角度

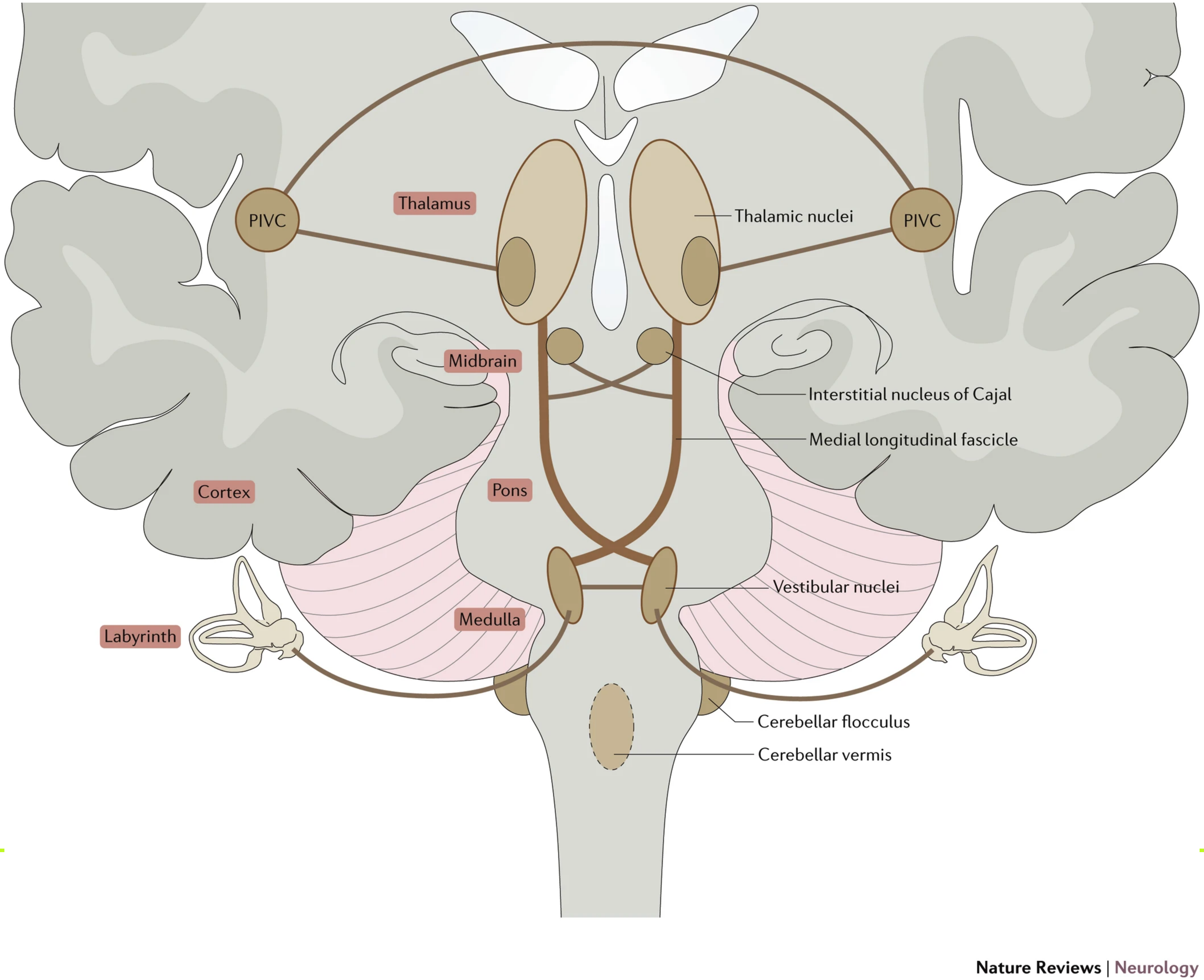
Ampulla Hair Cells





感受头旋转

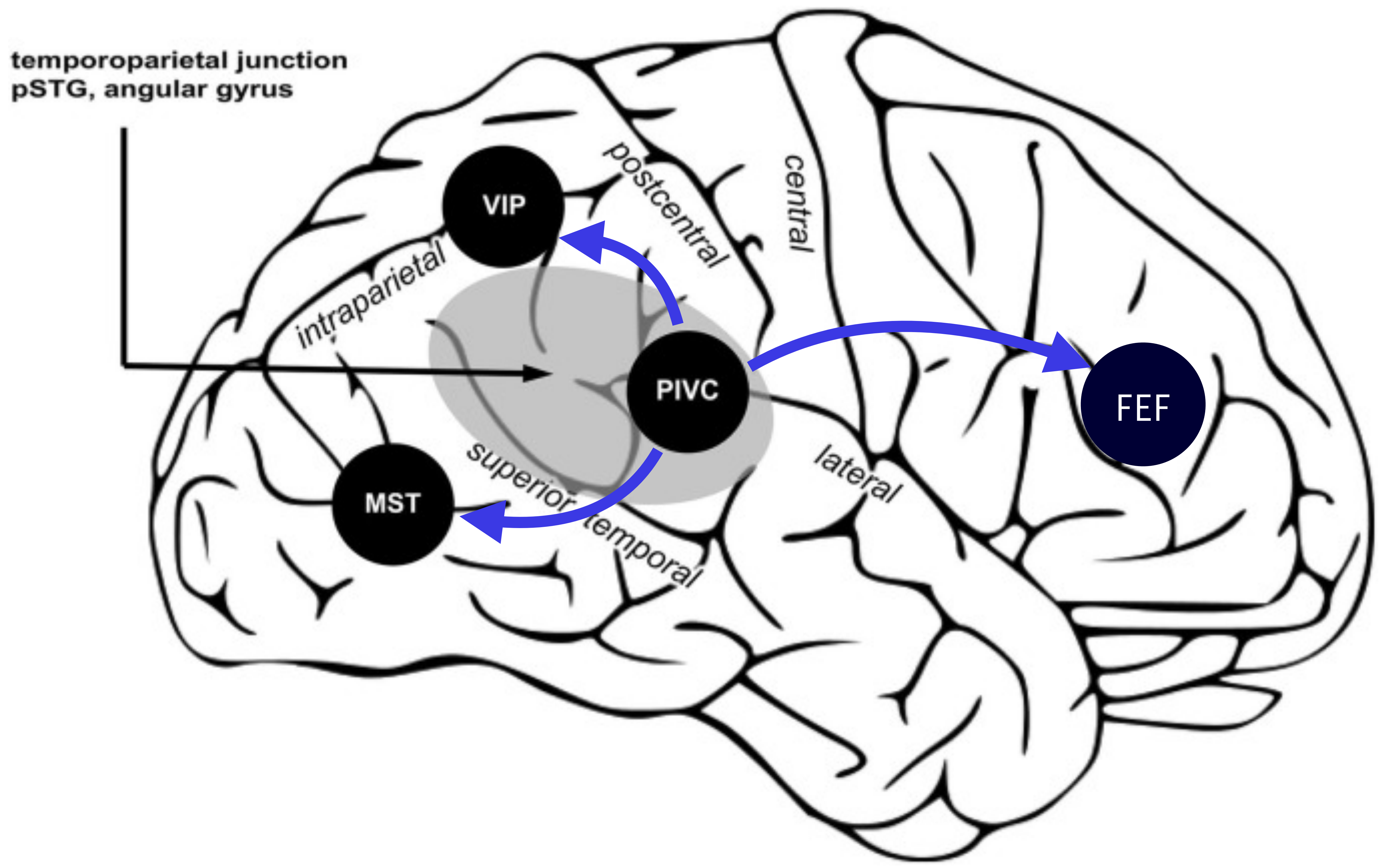
Central Vestibular Pathway



PIVC:

Parieto-insular vestibular cortex

Central Vestibular Pathway



- Auditory
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- **Somatosensory**
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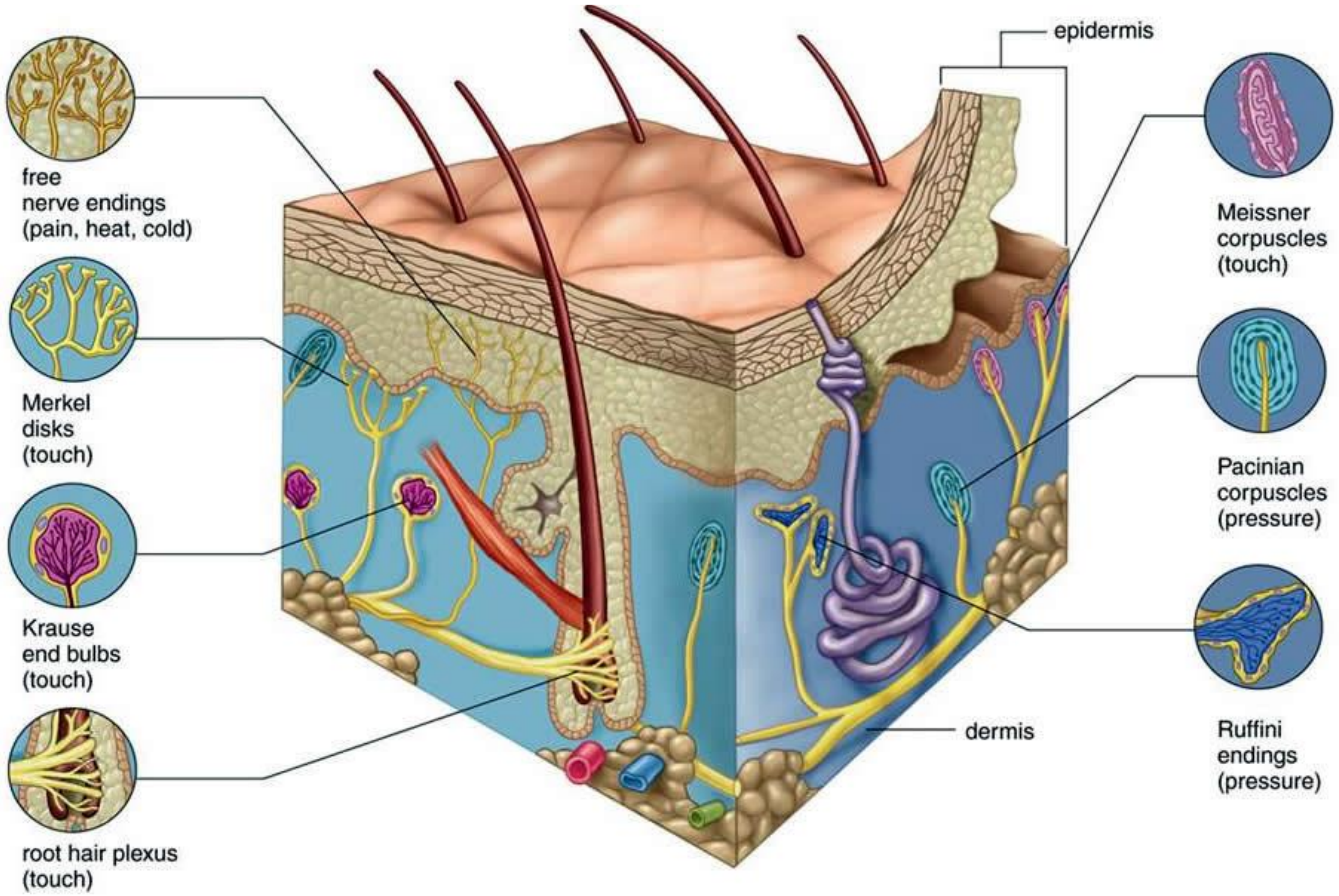
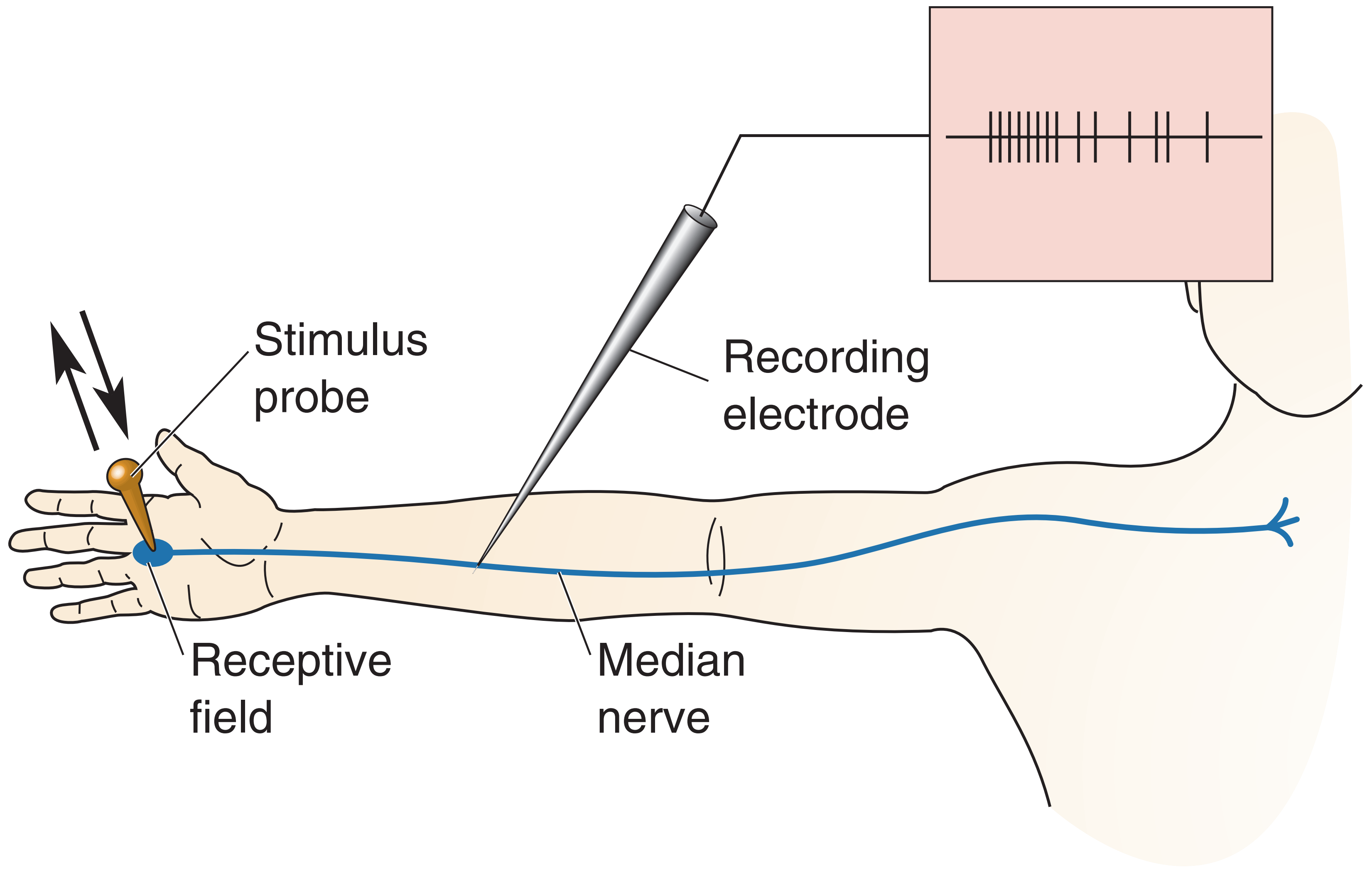
Many Types of Somatosensory Modalities

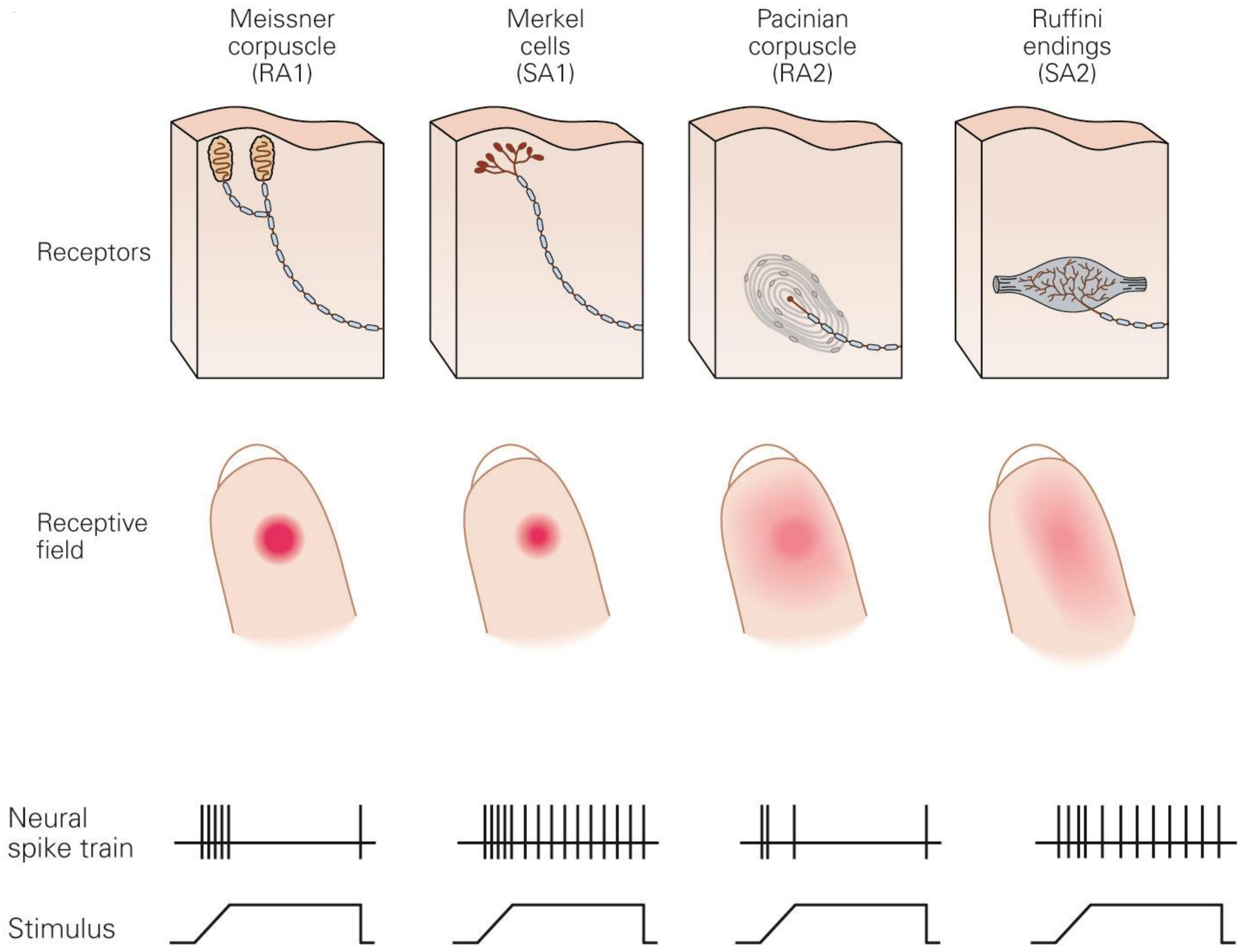


Somatosensory receptors are all over the body

There are many sub-modalities.

Basic Response Properties

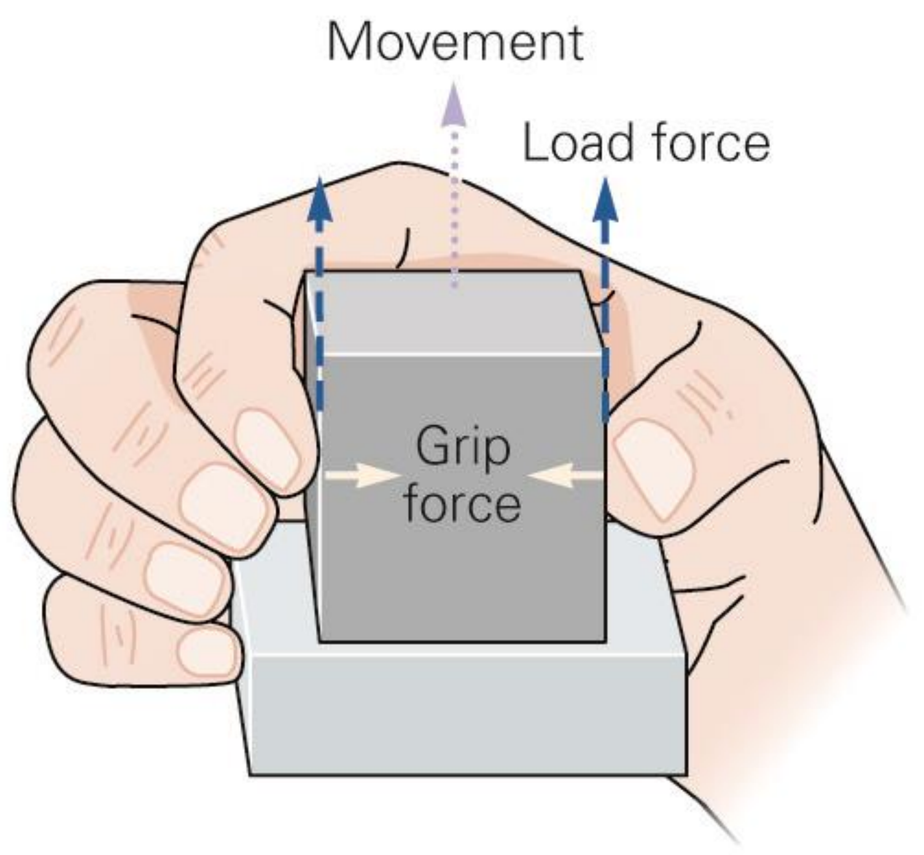


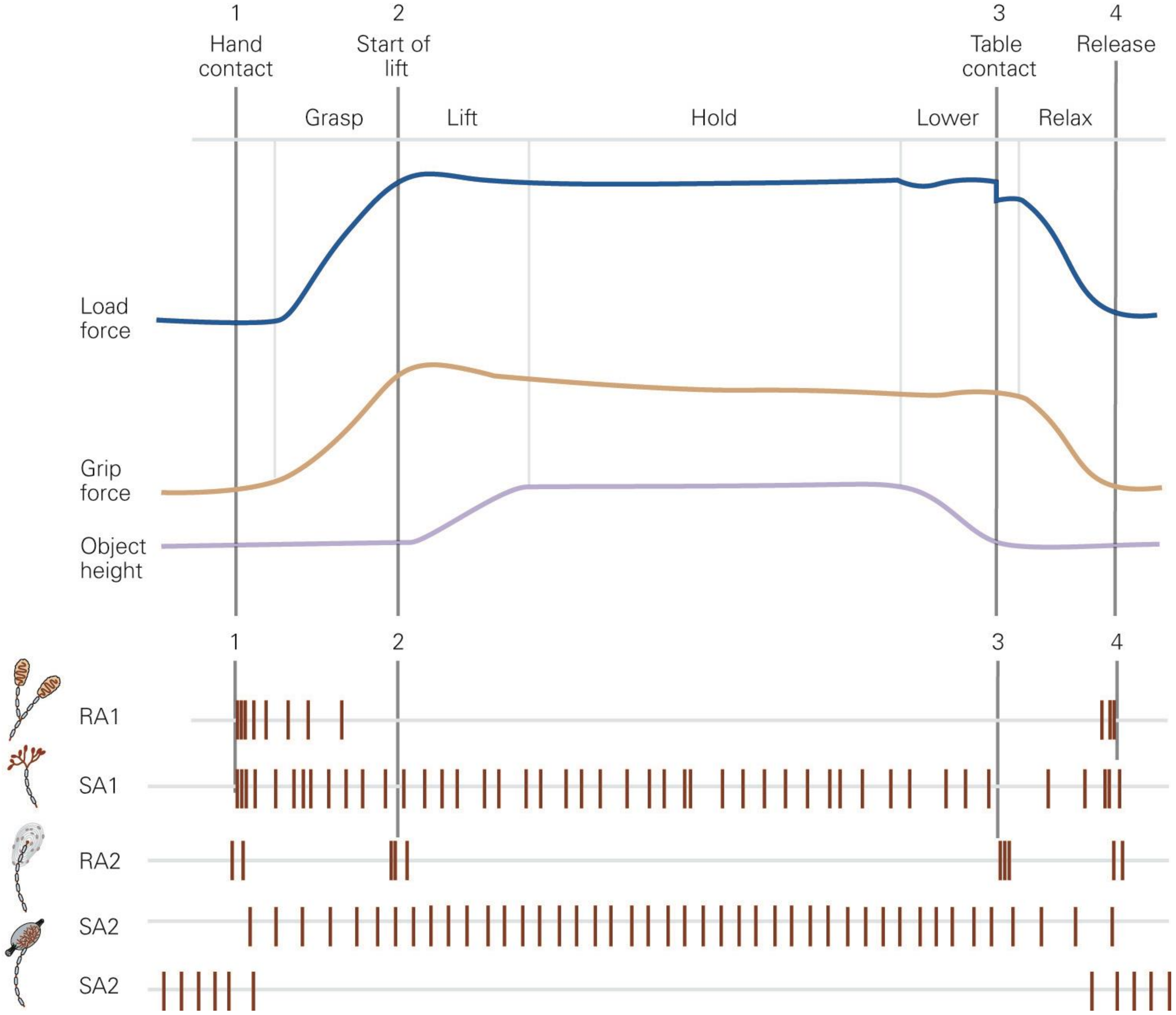


Adaptation	Rapid	Slow	Rapid	Slow
	RA1	SA1	RA2	SA2

- **Slowly adapting fibers detect object pressure and form**
- **Rapidly adapting fibers detect motion and vibration**

Working Together





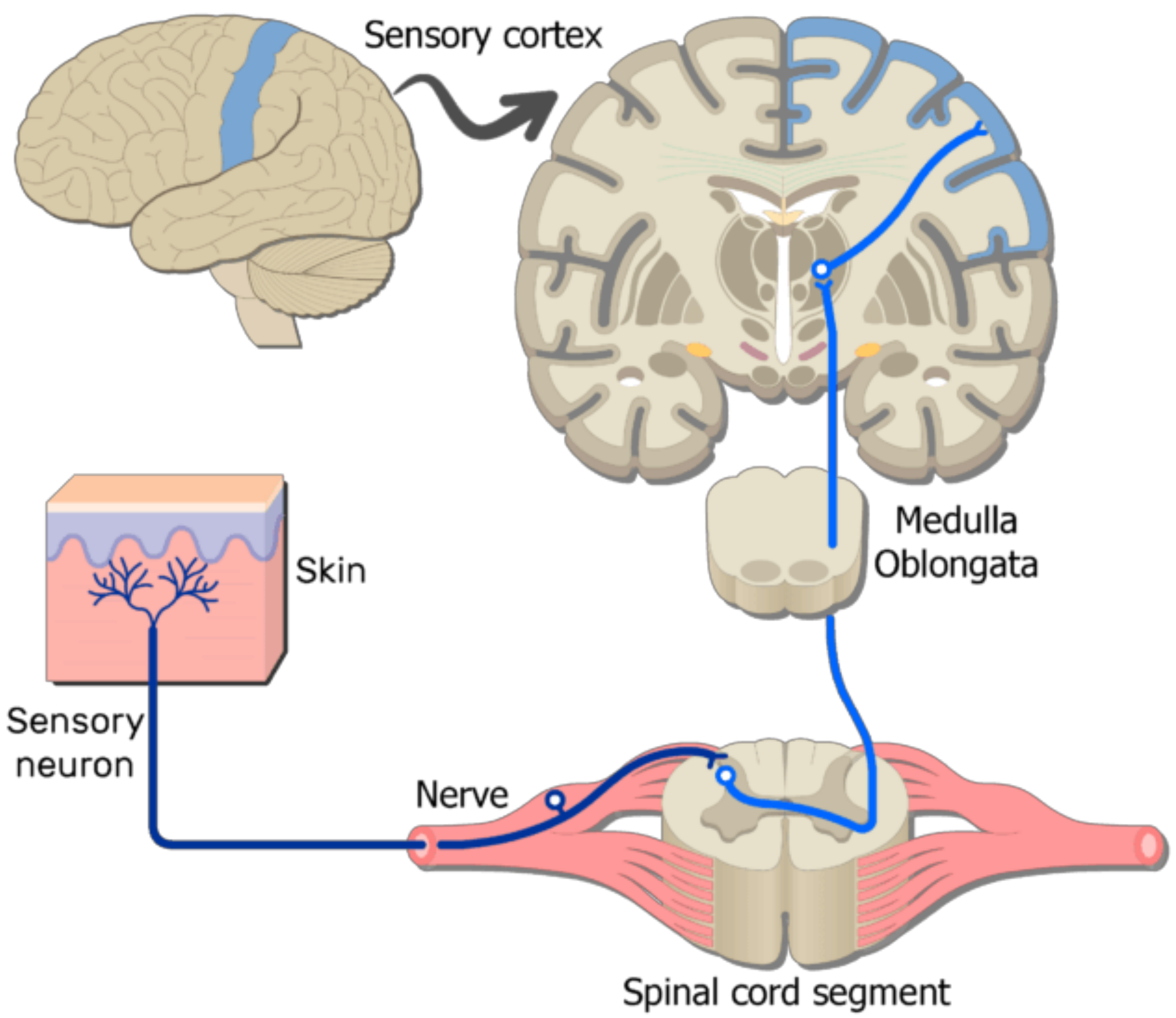
Touch

Grip force

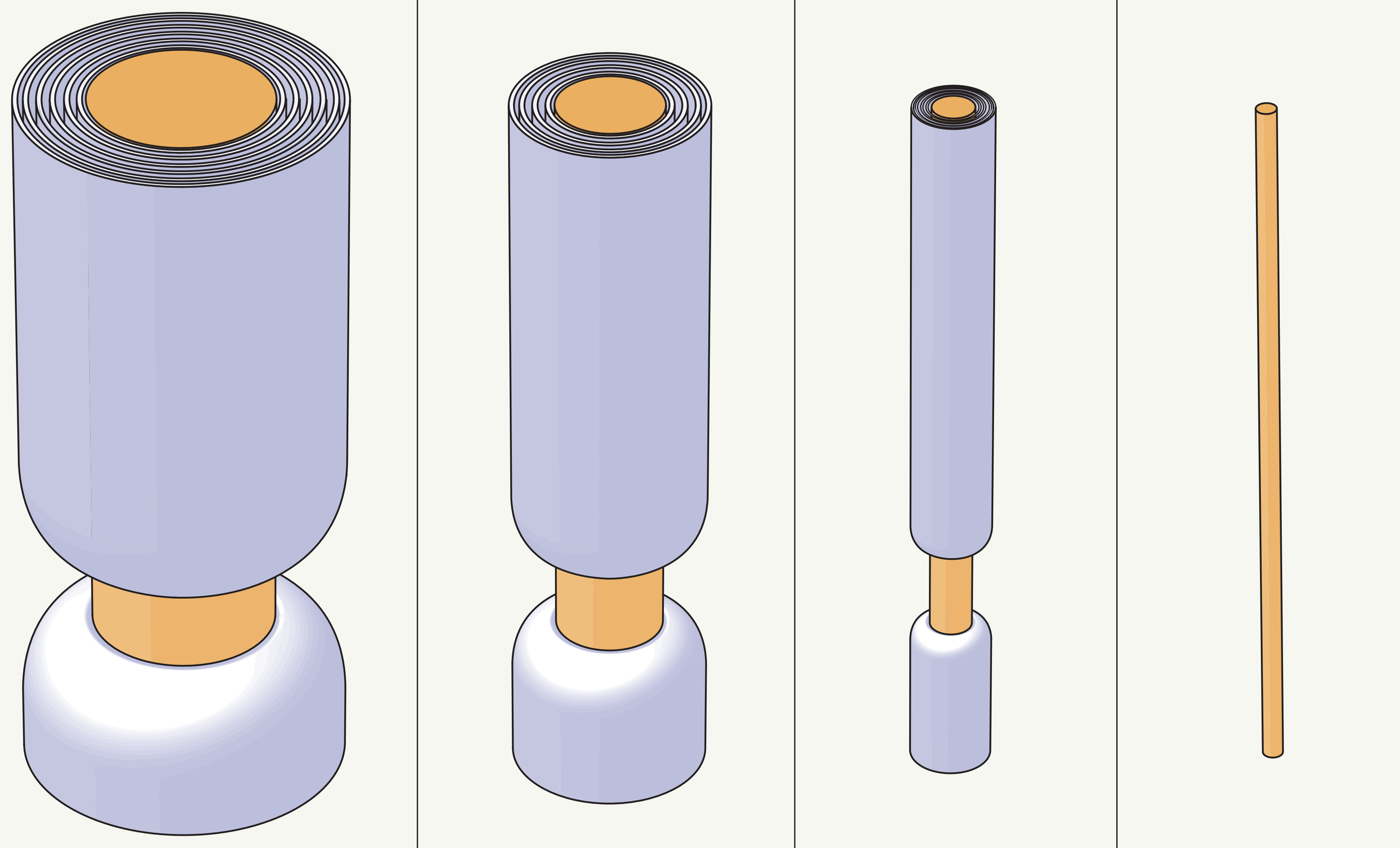
Vibrations

Hand Posture

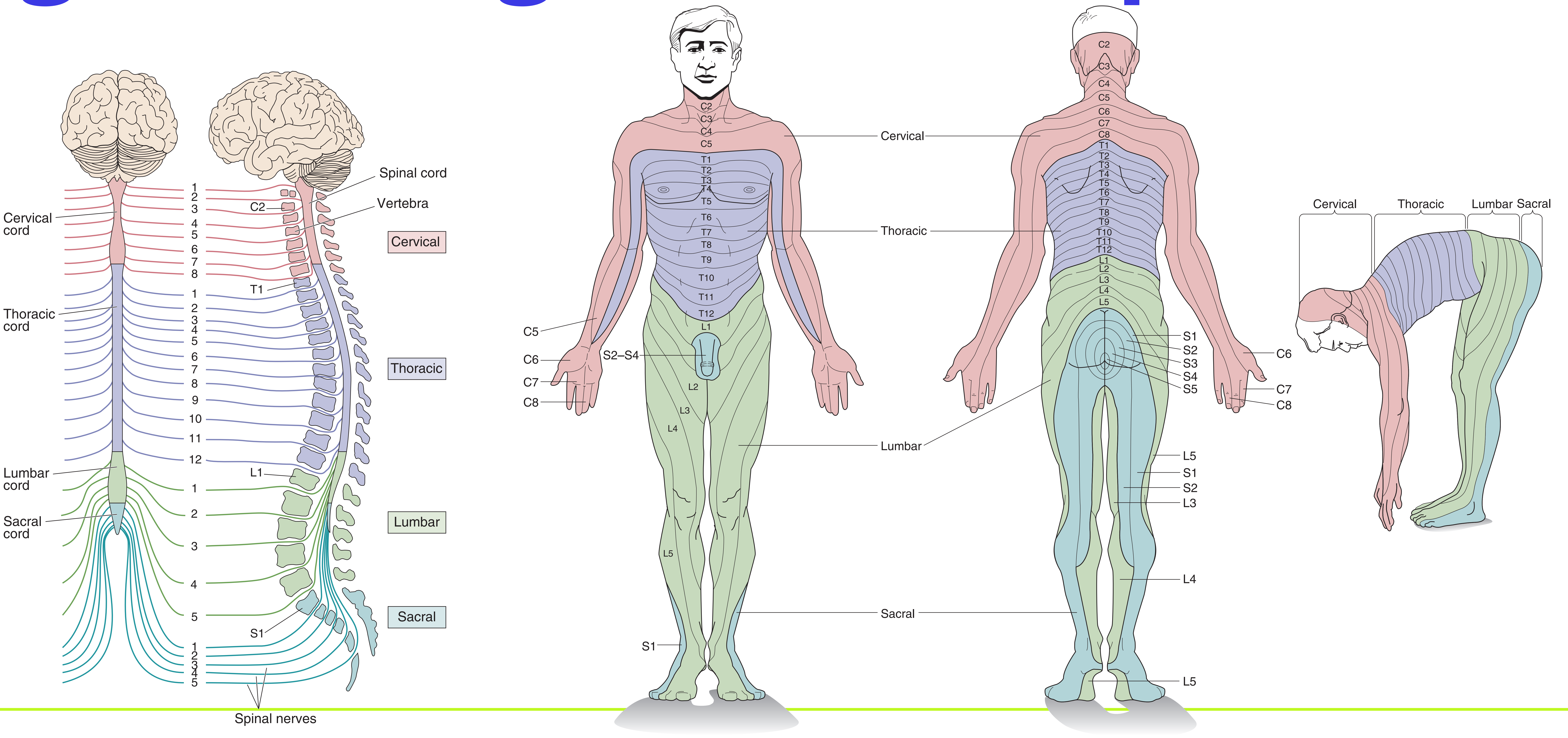
Somatosensory Pathway



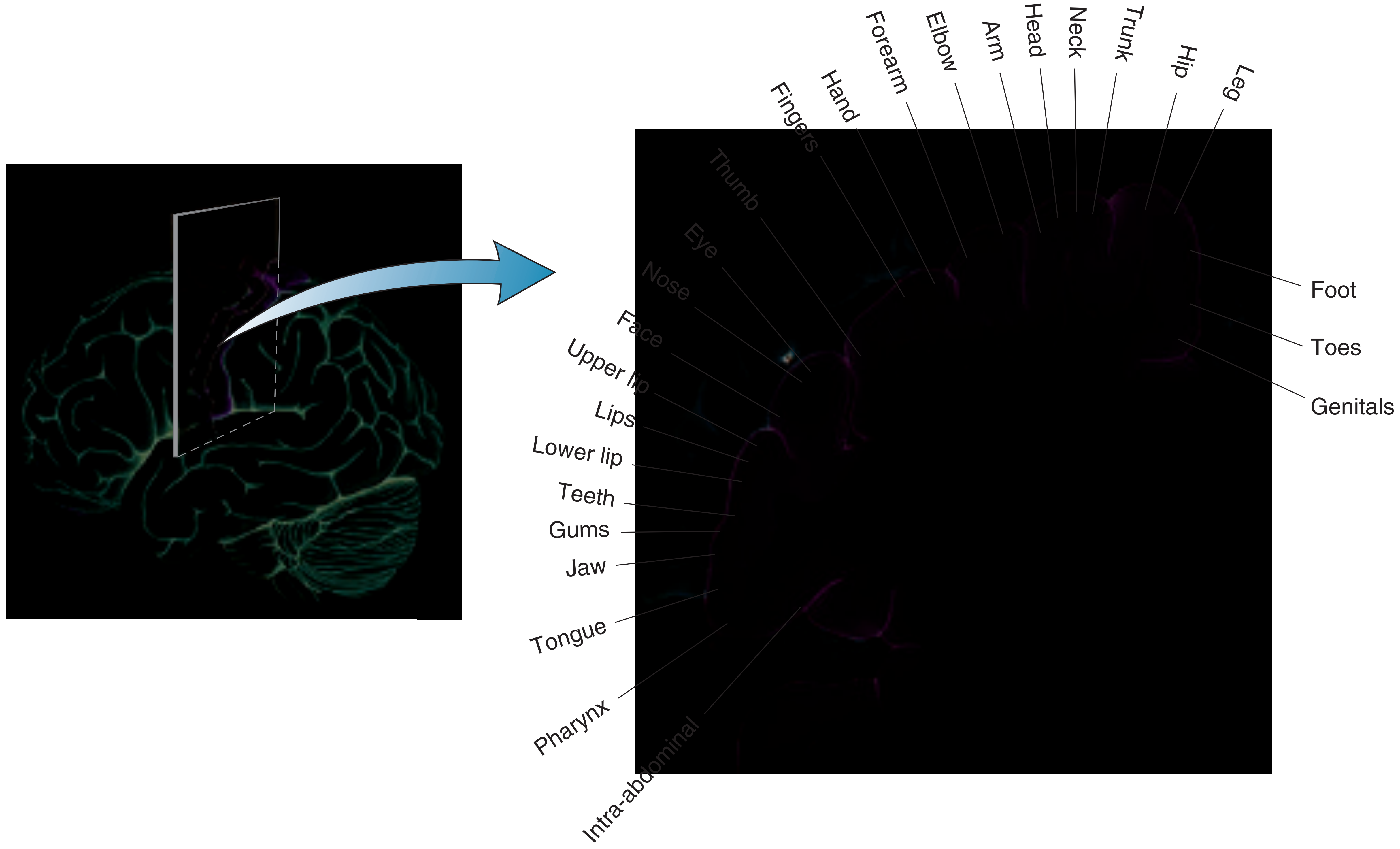
Axons from skin	Aα	Aβ	Aδ	C
Axons from muscles	Group I	II	III	IV
Diameter (μm)	13–20	6–12	1–5	0.2–1.5
Speed (m/sec)	80–120	35–75	5–30	0.5–2
Sensory receptors	Proprioceptors of skeletal muscle	Mechanoreceptors of skin	Pain, temperature	Temperature, pain, itch



Segmental Organization of Spinal Cord

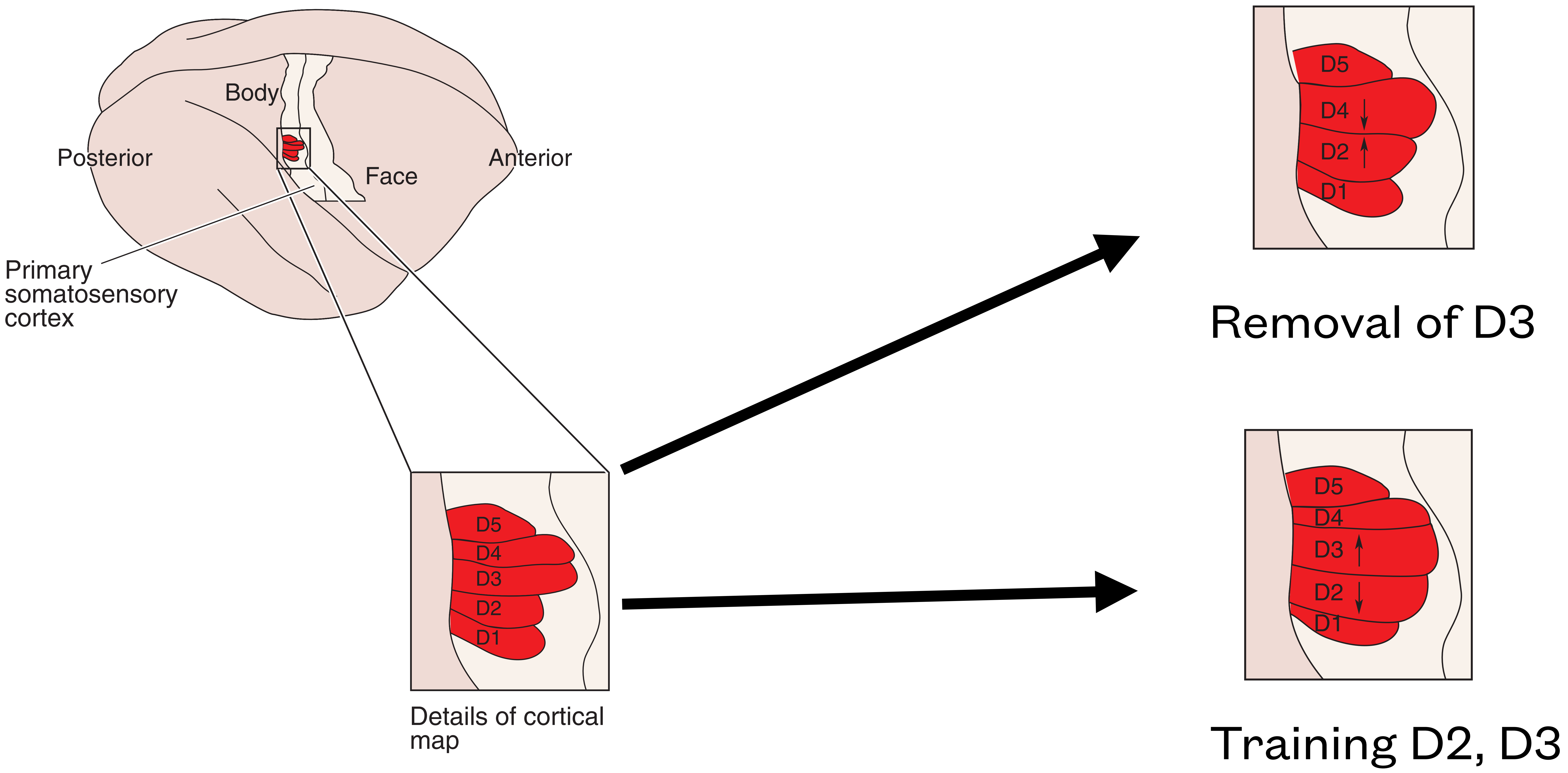


Somatotopic Map



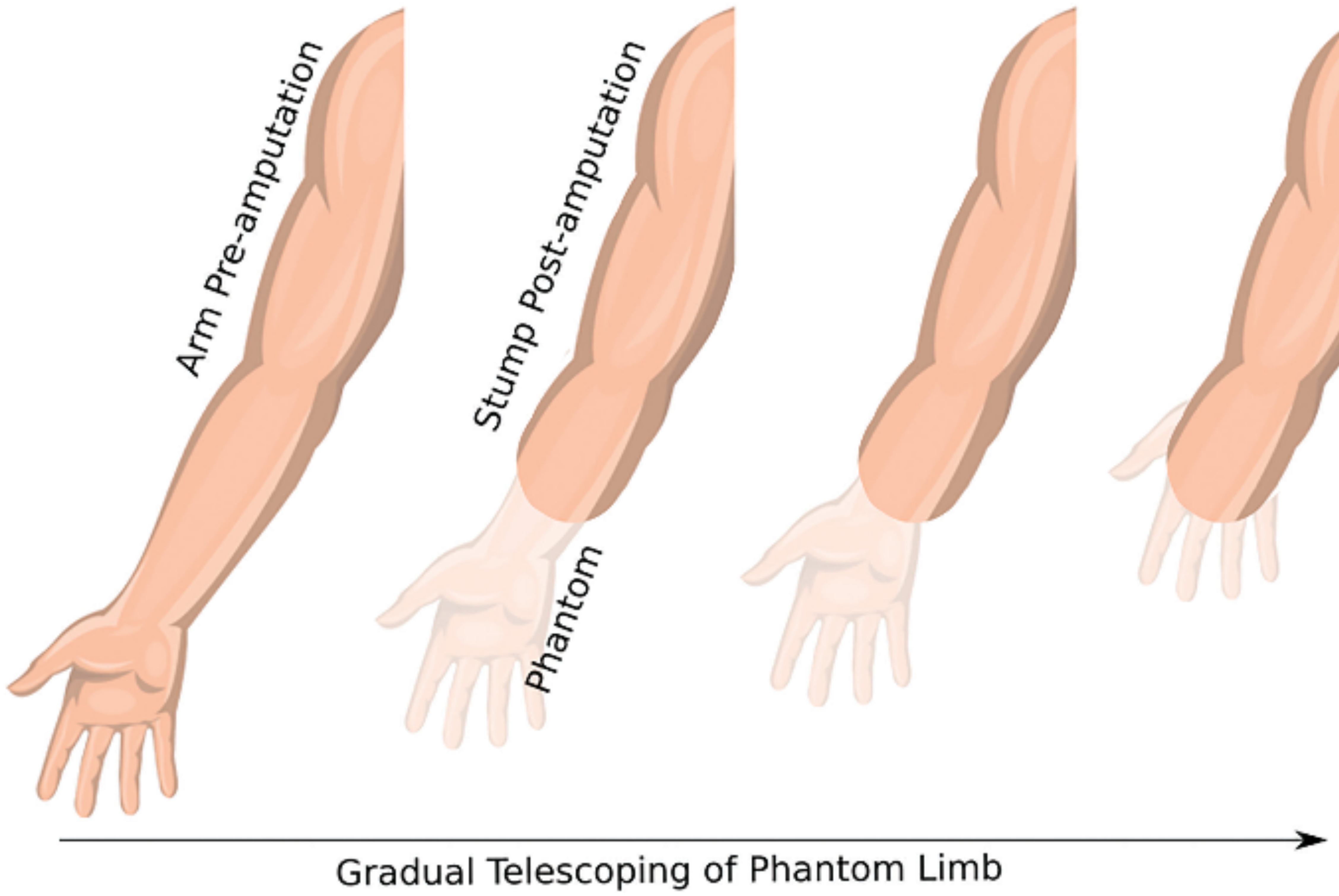
Homunculus

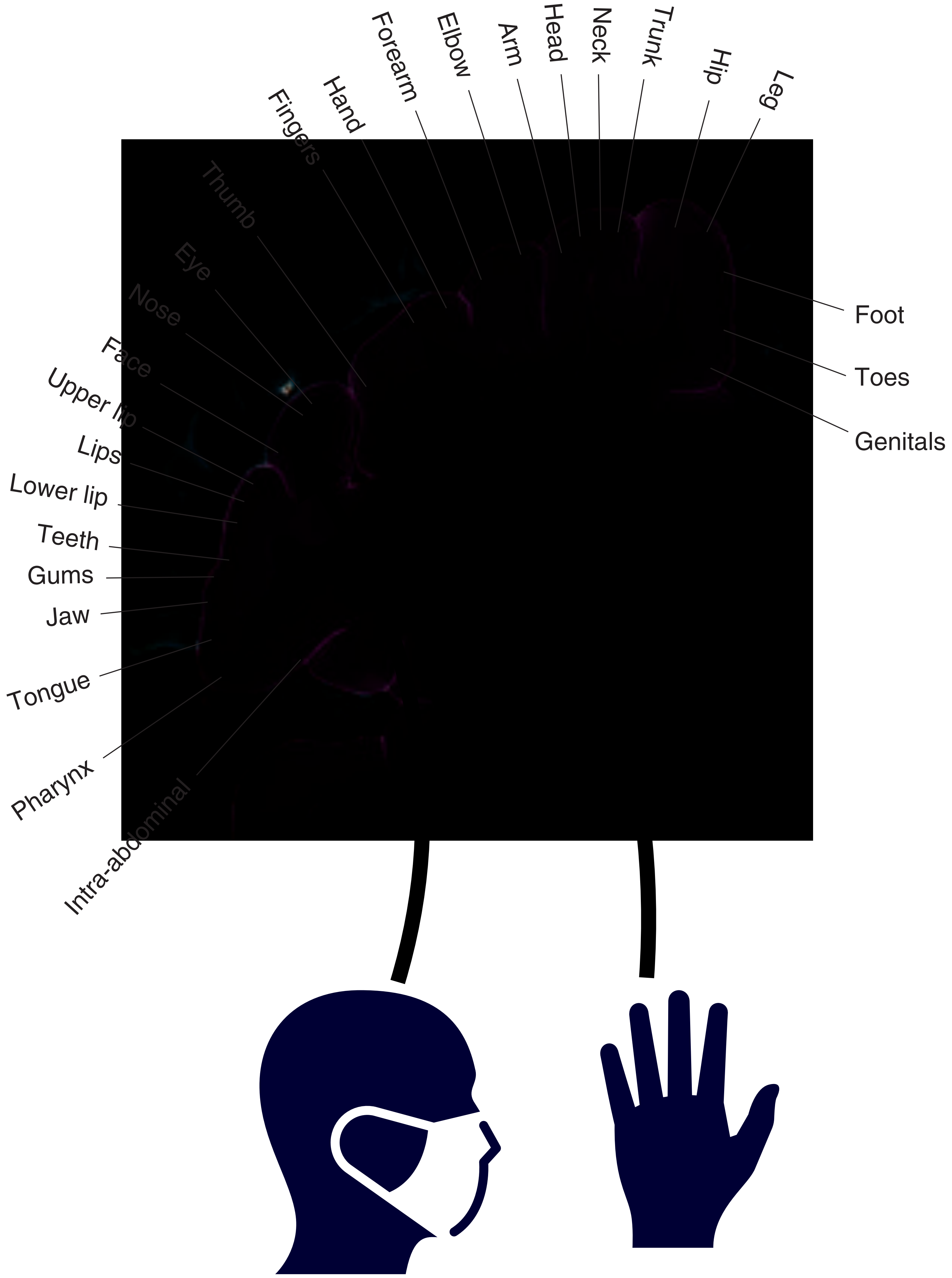
Plasticity



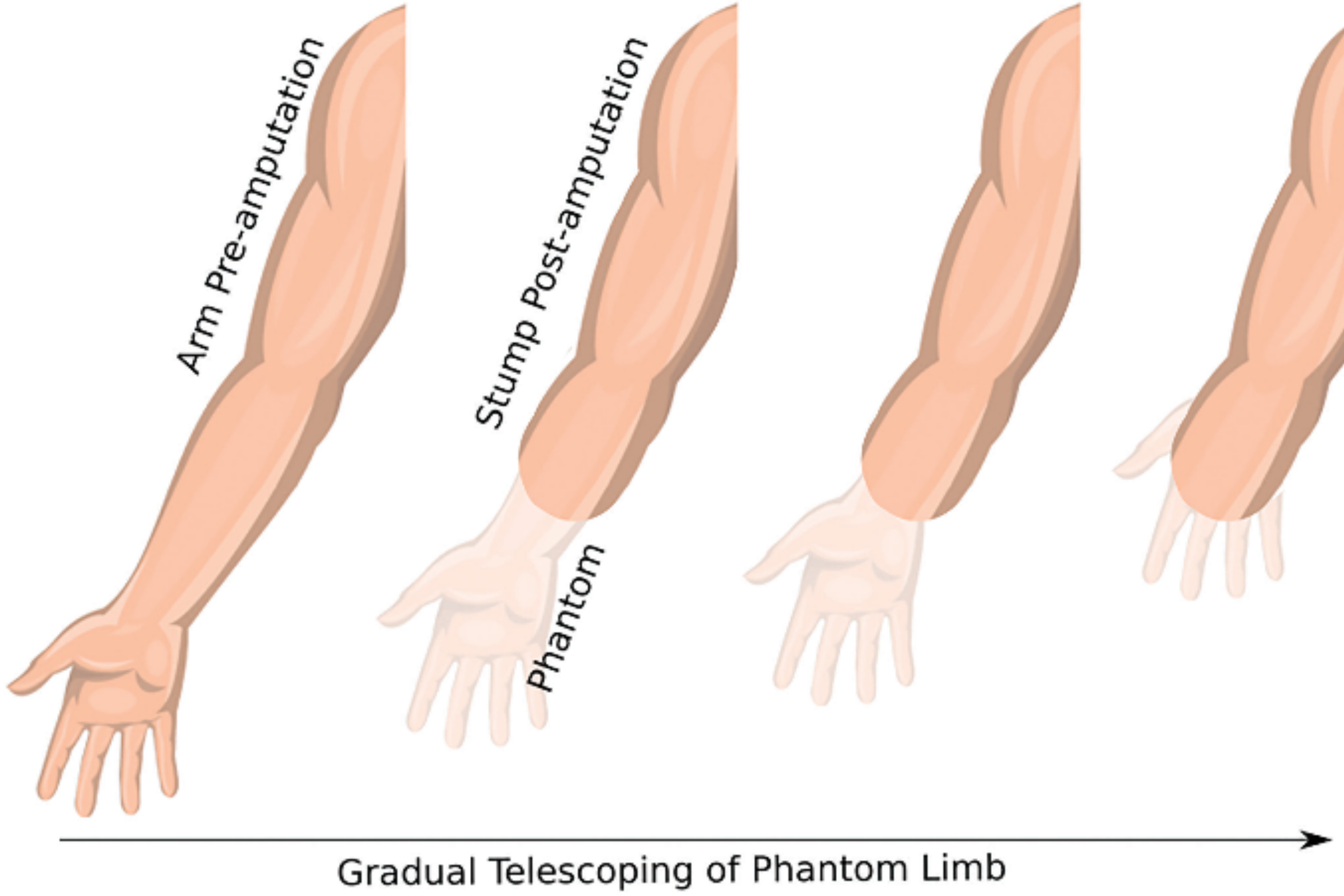
Body
Posterior
Face
Anterior
Primary somatosensory cortex
D5
D4
D3
D2
D1
Details of cortical map
D5
D4
D2
D1
Removal of D3
D5
D4
D3
D2
D1
Training D2, D3

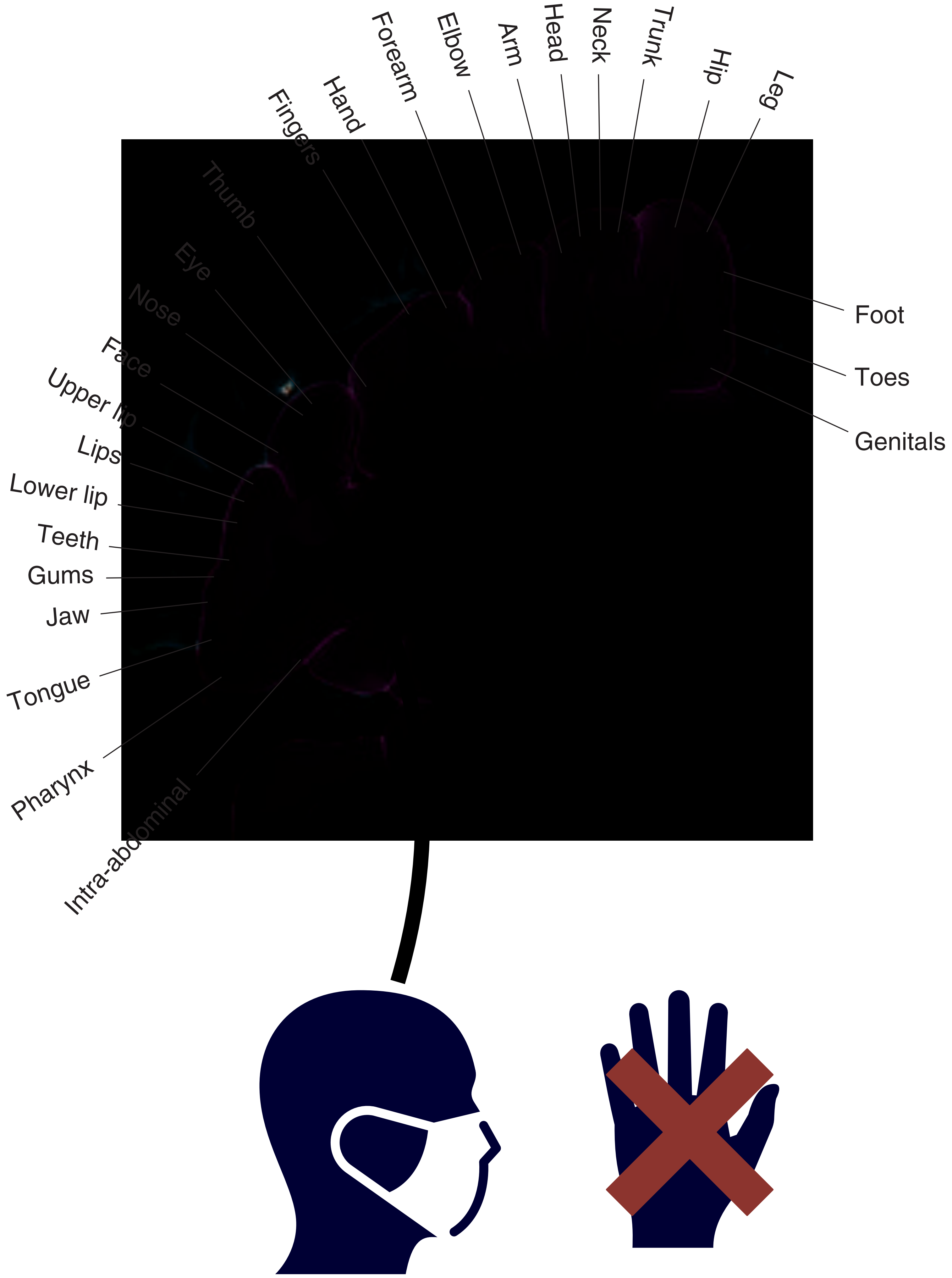
Phantom Limb





Phantom Limb





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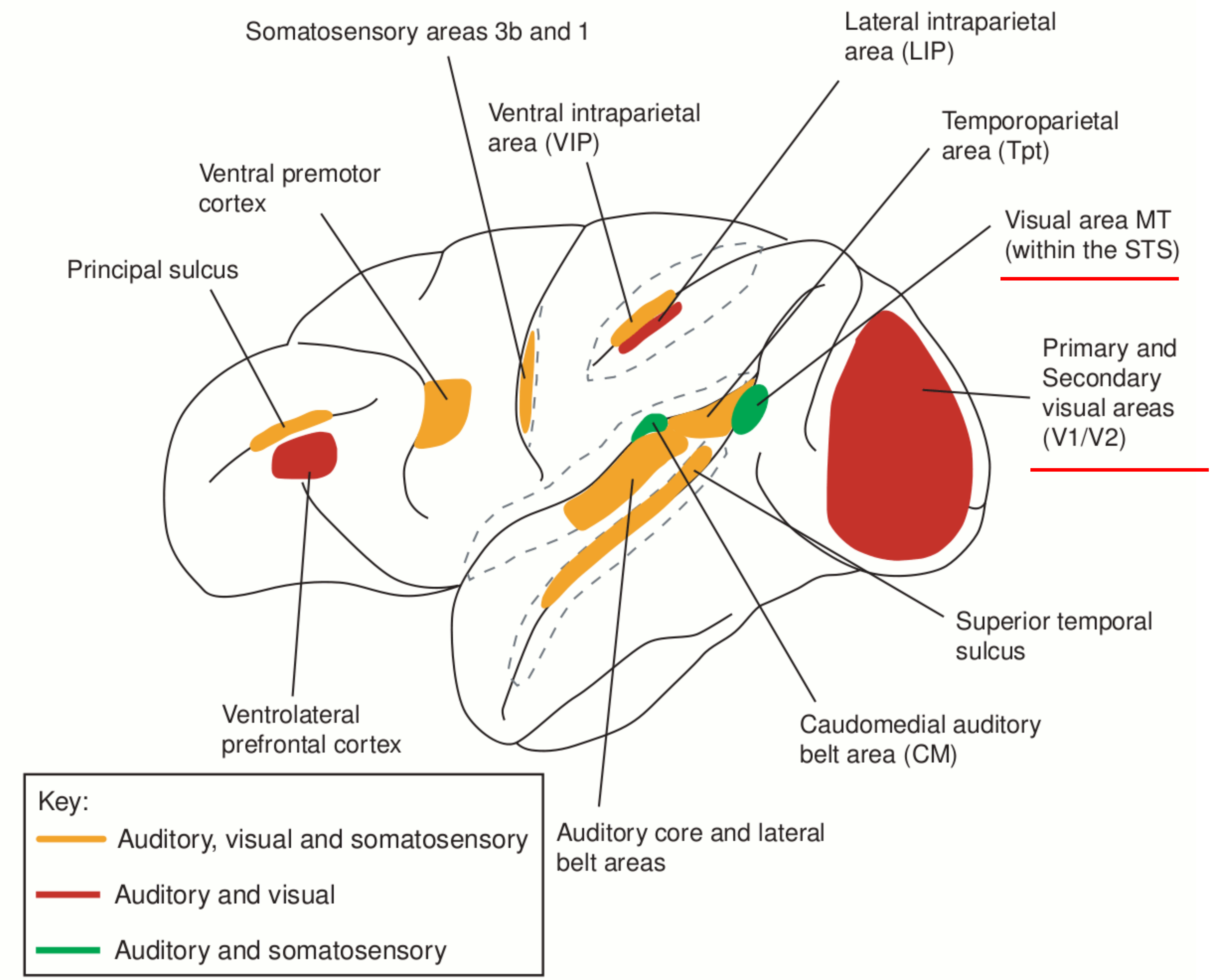
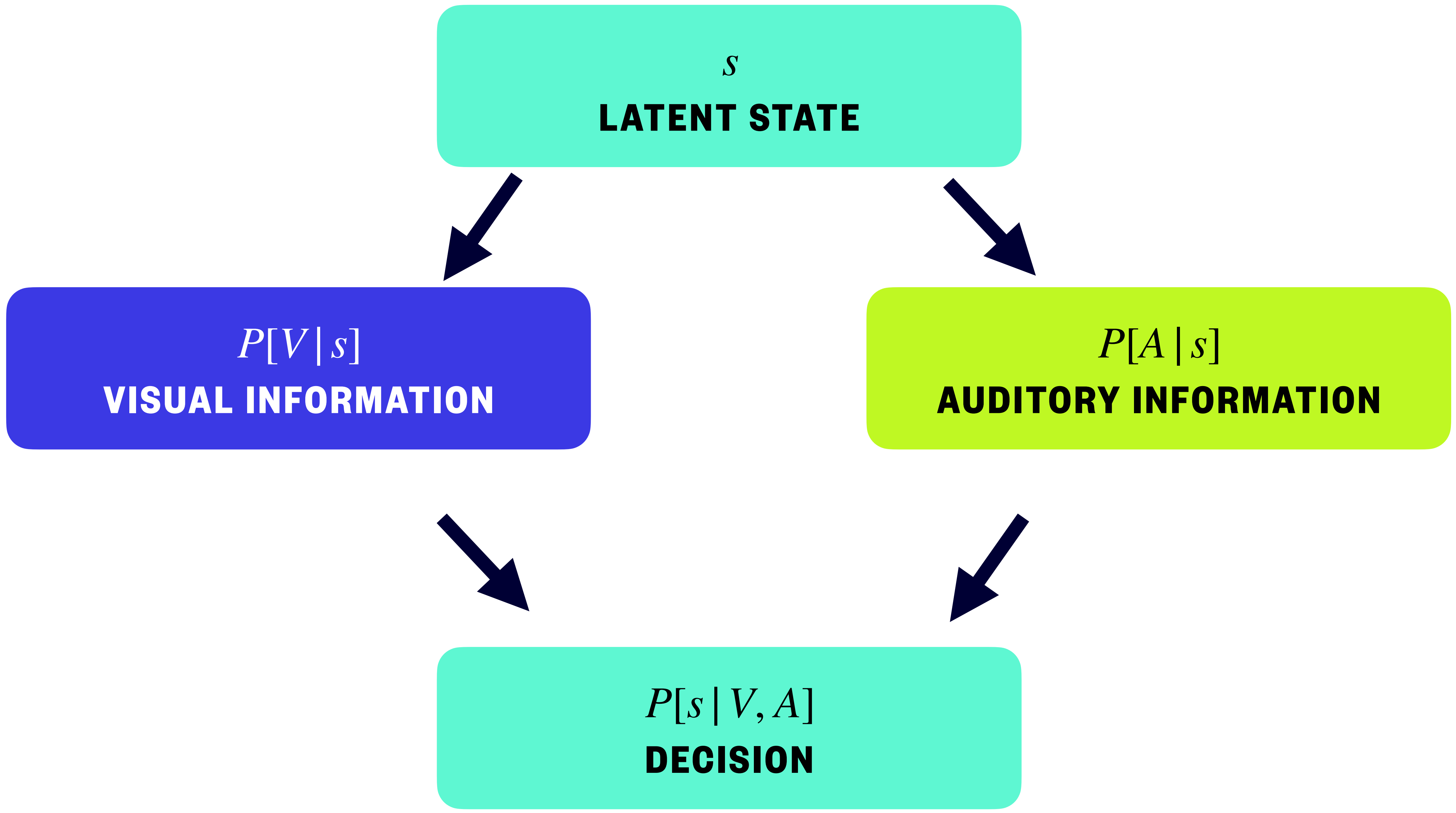
Multimodal Brain



Linear Optimal Multi-Modality Integration



Bayesian Inference

s
LATENT STATE

$P[V \mid s]$
VISUAL INFORMATION

$P[A \mid s]$
AUDITORY INFORMATION

$$P(r_V \mid s) = \frac{1}{\sqrt{2\pi\sigma_V^2}} e^{-\frac{(r_V - s)^2}{2\sigma_V^2}}$$

$$P(r_A \mid s) = \frac{1}{\sqrt{2\pi\sigma_A^2}} e^{-\frac{(r_A - s)^2}{2\sigma_A^2}}$$

$P[s \mid V, A]$
DECISION

$$P(s \mid r_A, r_V) = \frac{P(r_A, r_V \mid s)P(s)}{P(r_A, r_V)} \propto P(r_A \mid s)P(r_V \mid s)$$

$$\propto e^{-\frac{(r_V - s)^2}{2\sigma_V^2}} \bullet e^{-\frac{(r_A - s)^2}{2\sigma_A^2}} = e^{-\frac{1}{2}[\frac{(r_V - s)^2}{\sigma_V^2} + \frac{(r_A - s)^2}{\sigma_A^2}]}$$

Maximum A Posteriori (MAP) Estimate

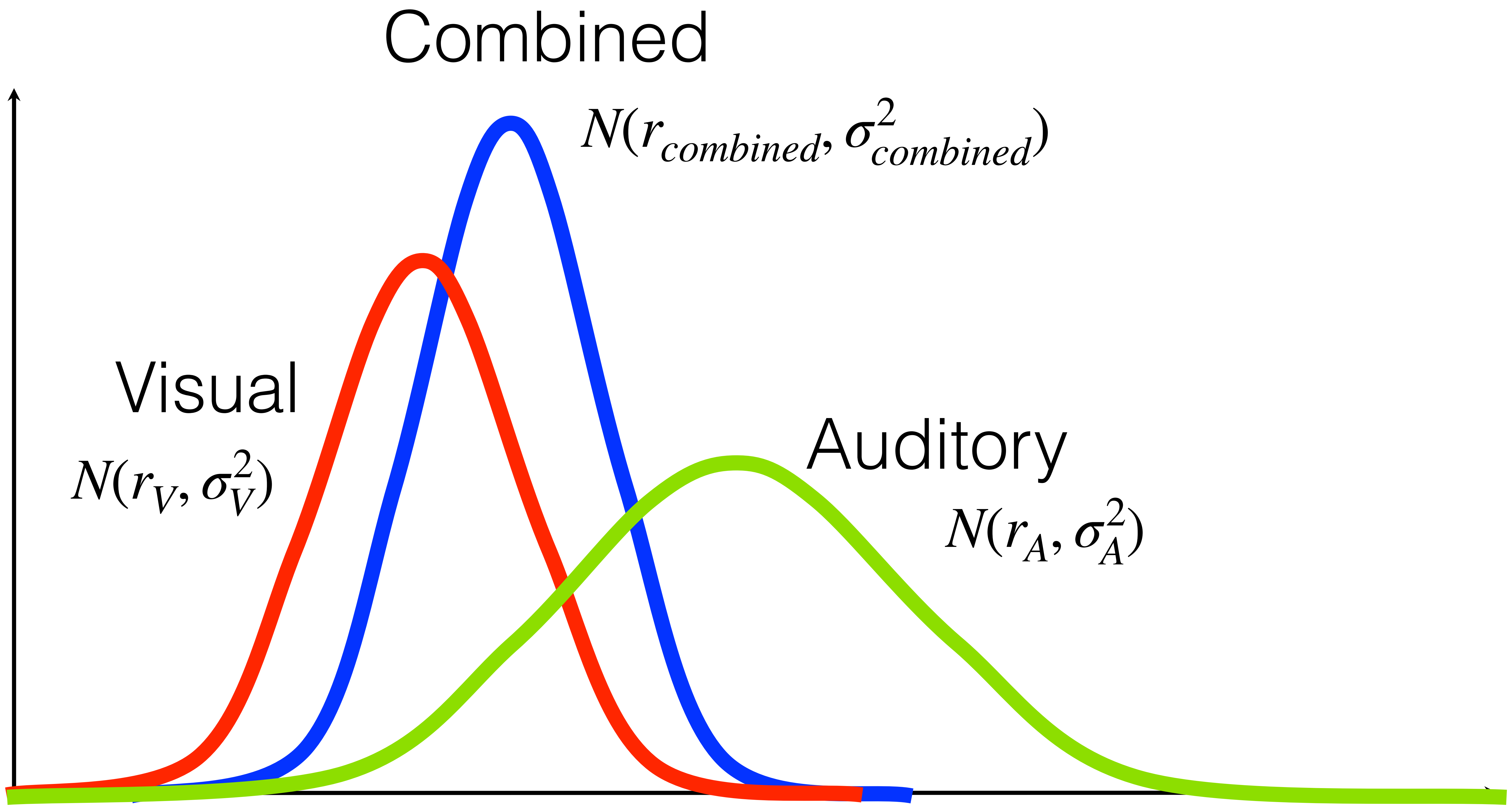
$$\underset{s}{\operatorname{argmax}}\, P(s \,|\, r_A, r_V) = e^{-\frac{1}{2}[\frac{(r_V - s)^2}{\sigma_V^2} + \frac{(r_A - s)^2}{\sigma_A^2}]}$$

$$\frac{\partial}{\partial s}(\frac{(r_V - s)^2}{\sigma_V^2} + \frac{(r_A - s)^2}{\sigma_A^2}) = 0$$

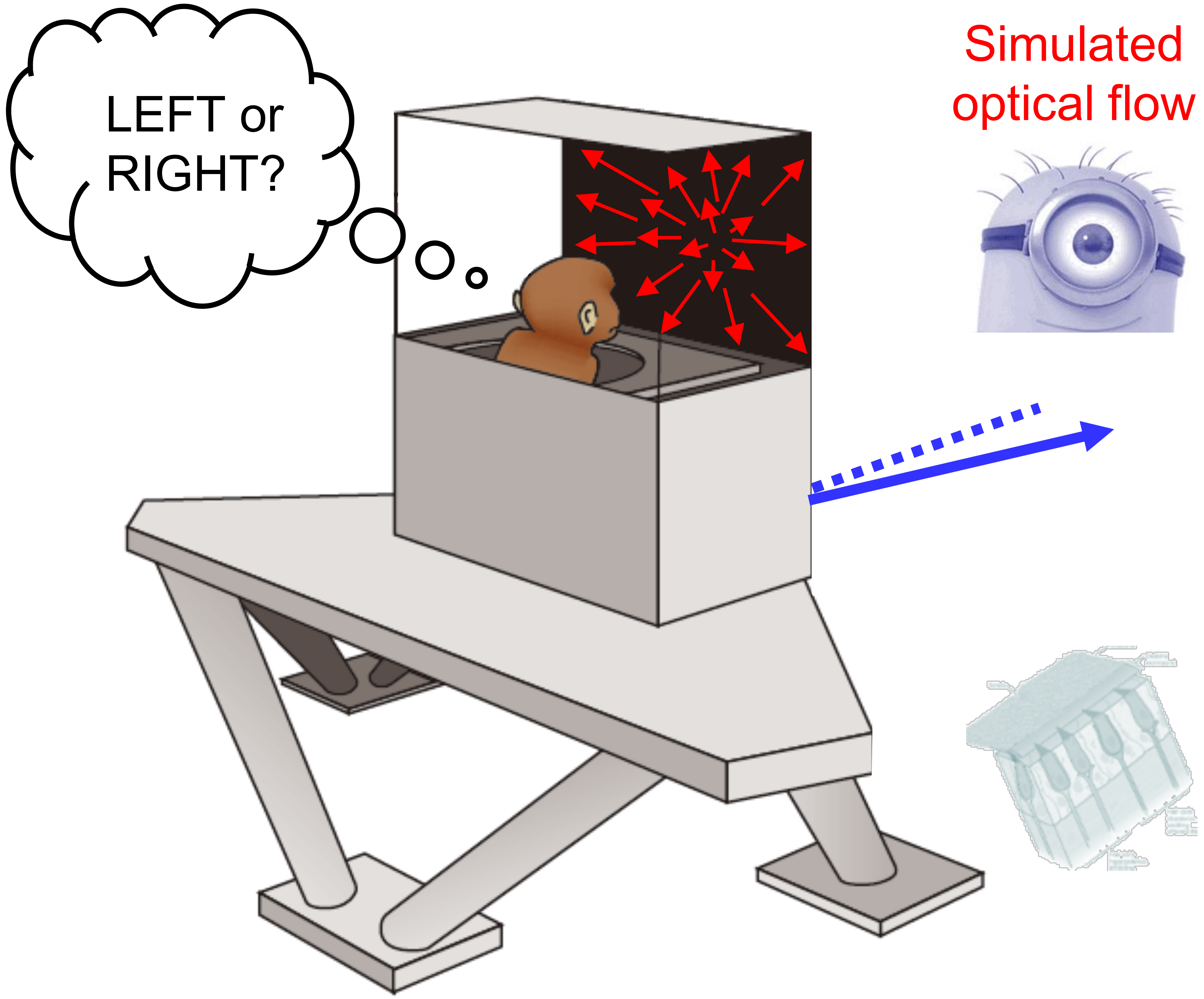
$$-2(\frac{r_V - s}{\sigma_V^2} + \frac{r_A - s}{\sigma_A^2}) = 0$$

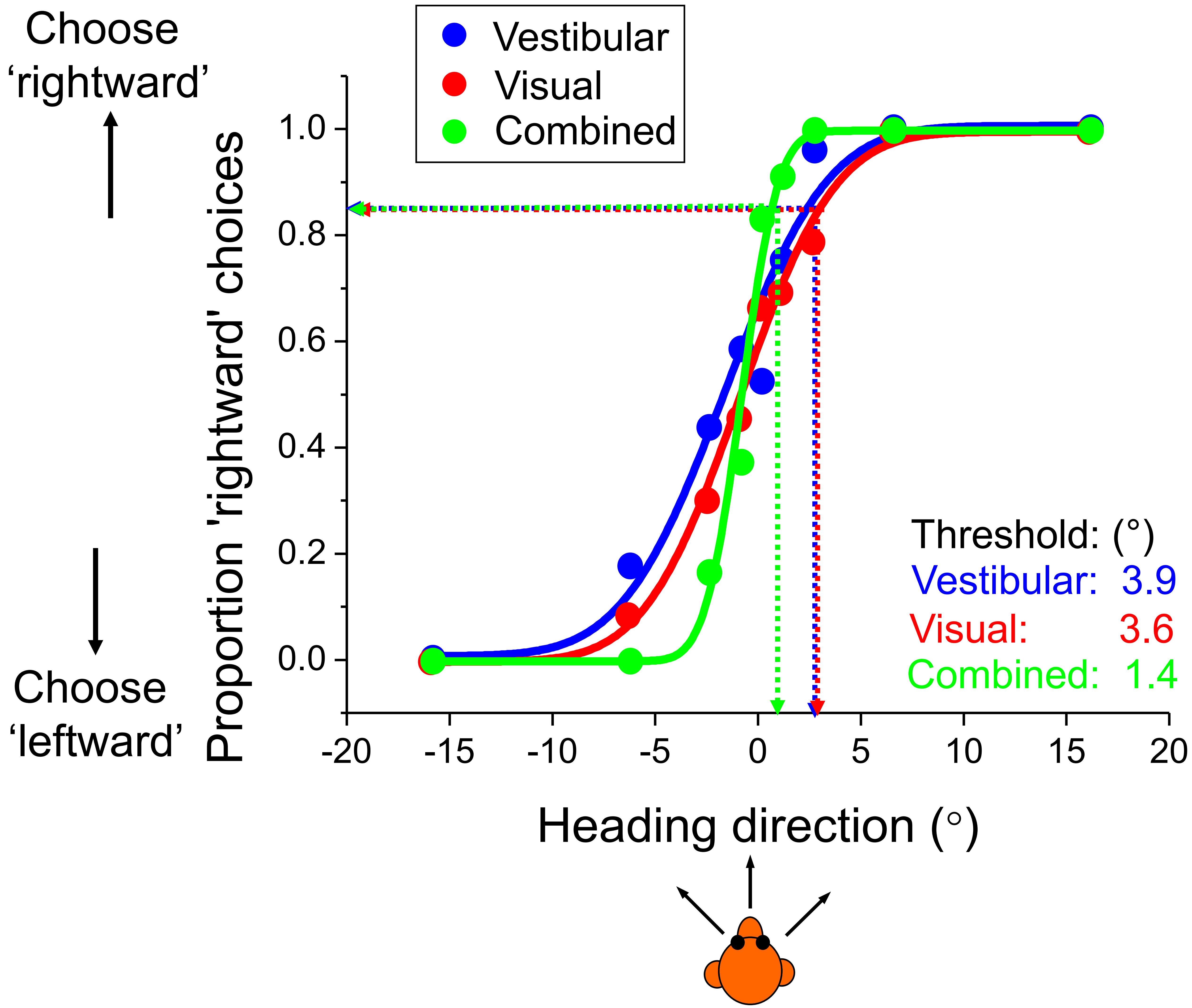
$$s = \frac{\frac{r_v}{\sigma_v^2} + \frac{r_A}{\sigma_A^2}}{\frac{1}{\sigma_v^2} + \frac{1}{\sigma_A^2}}$$

Linear Optimal Multi-Modality Integration

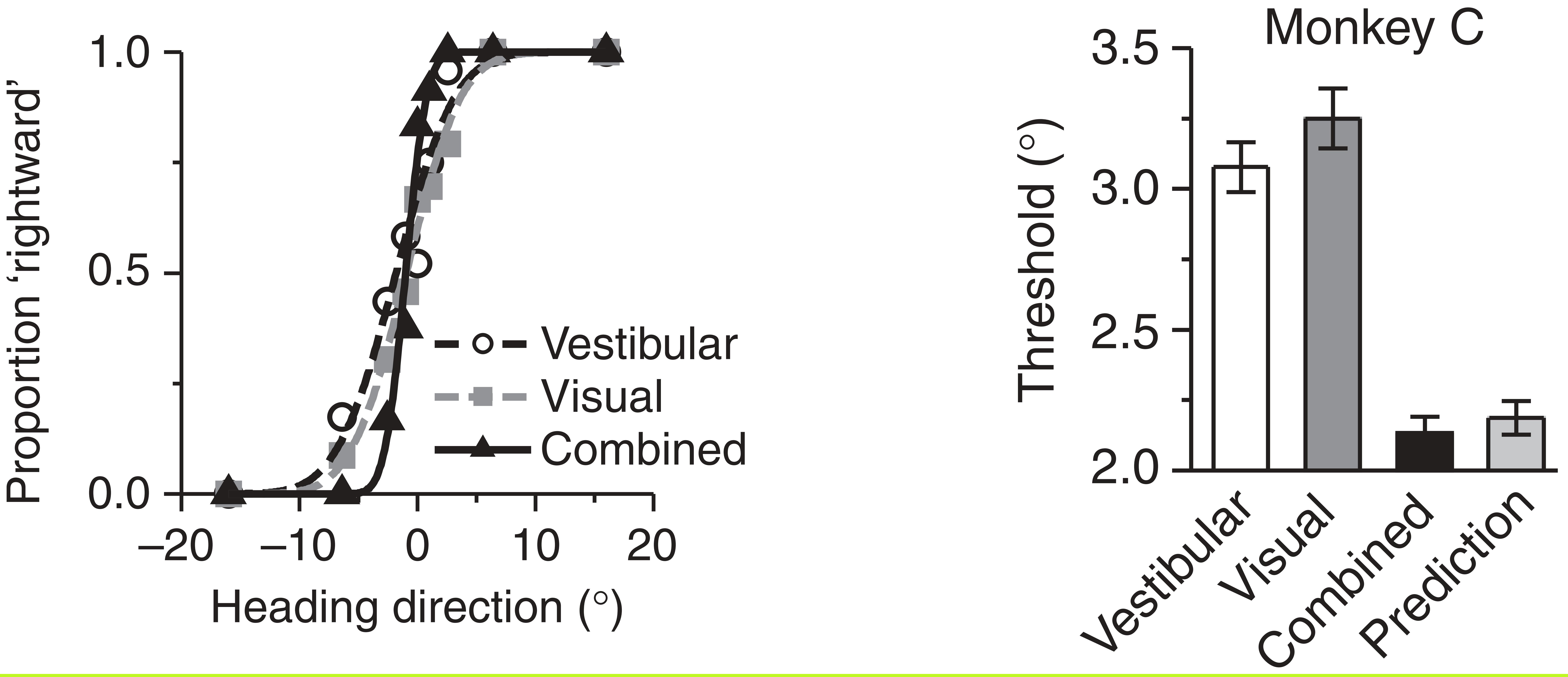


Experiments

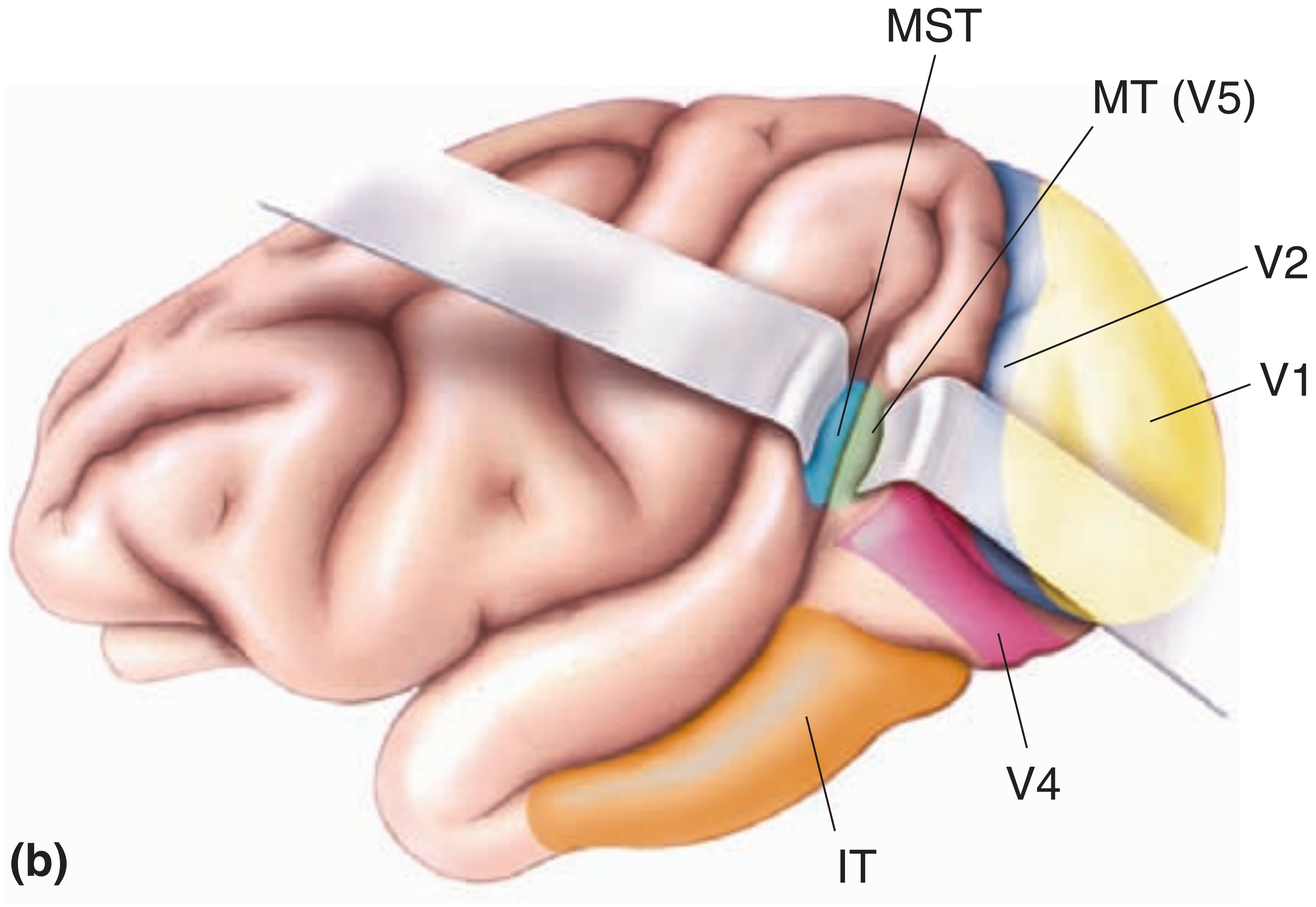


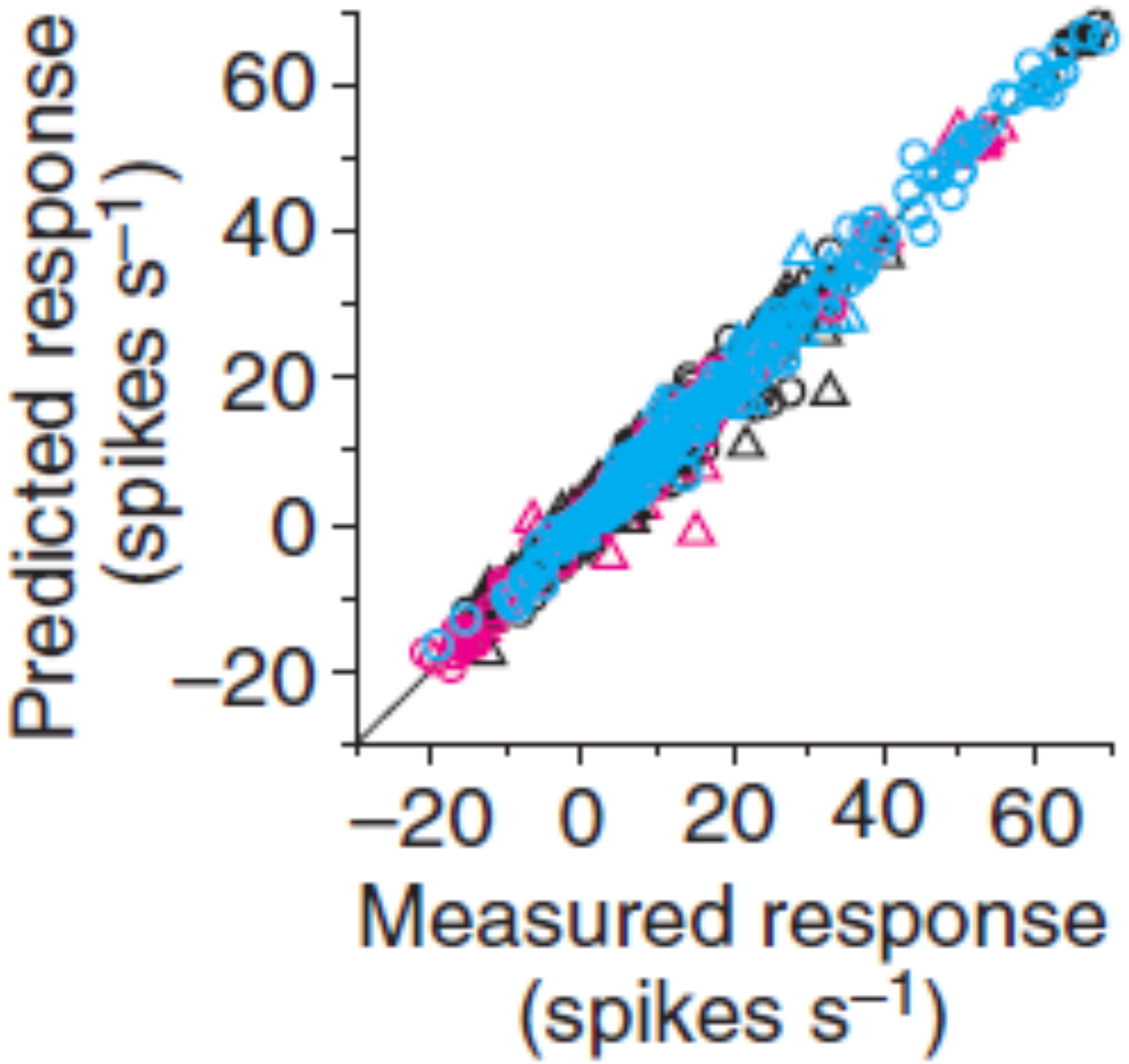


Dora Angelaki, Greg Deangelis, 顾勇


Proportion 'rightward'
1.0
0.5
0.0
–20
–10
0
10
20
Heading direction (°)
Vestibular
Visual
Combined
Monkey C
Threshold (°)
3.5
3.0
2.5
2.0
Vestibular
Visual
Combined
Prediction

MST Neurons Combined Information





Gu et al., 2008

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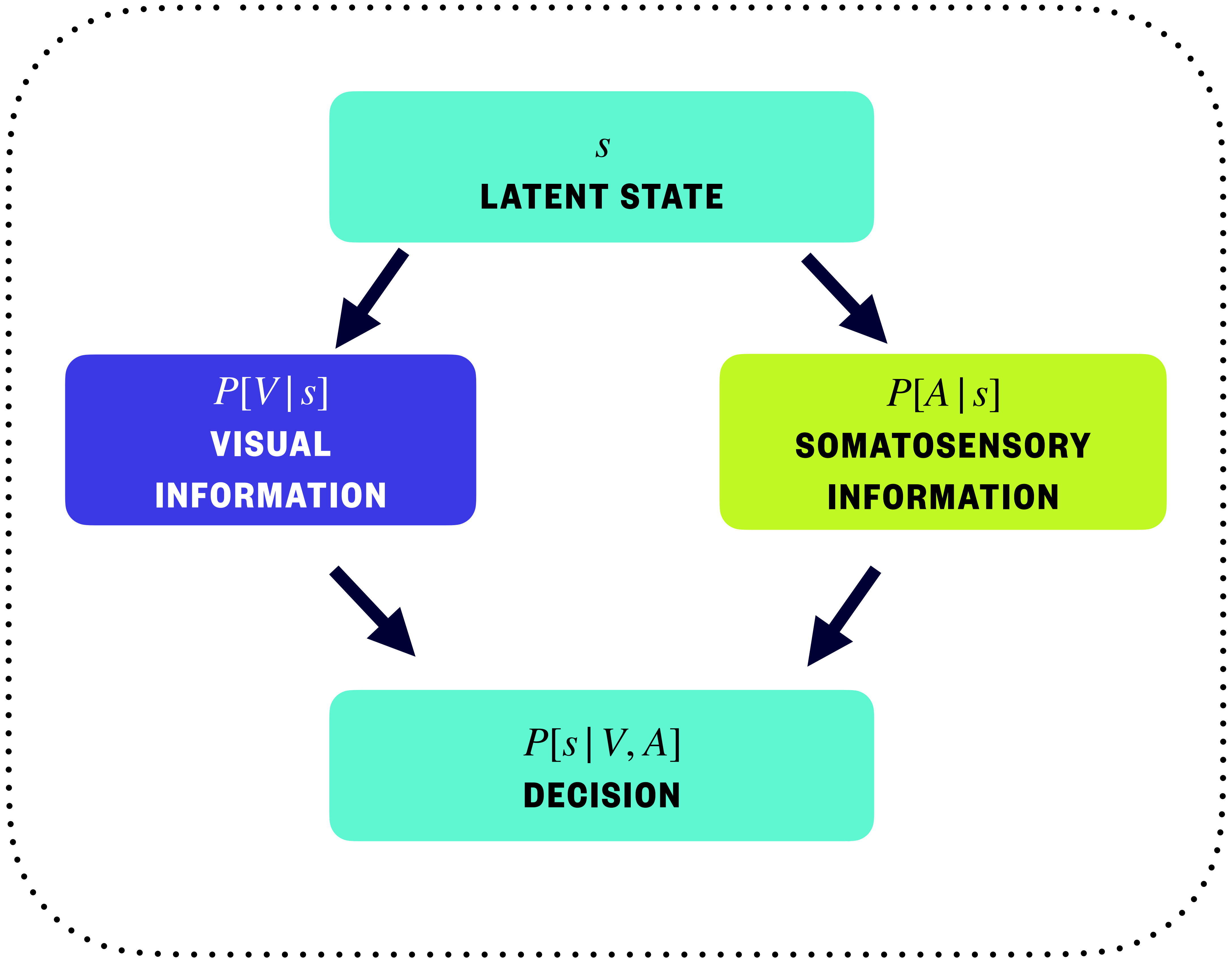
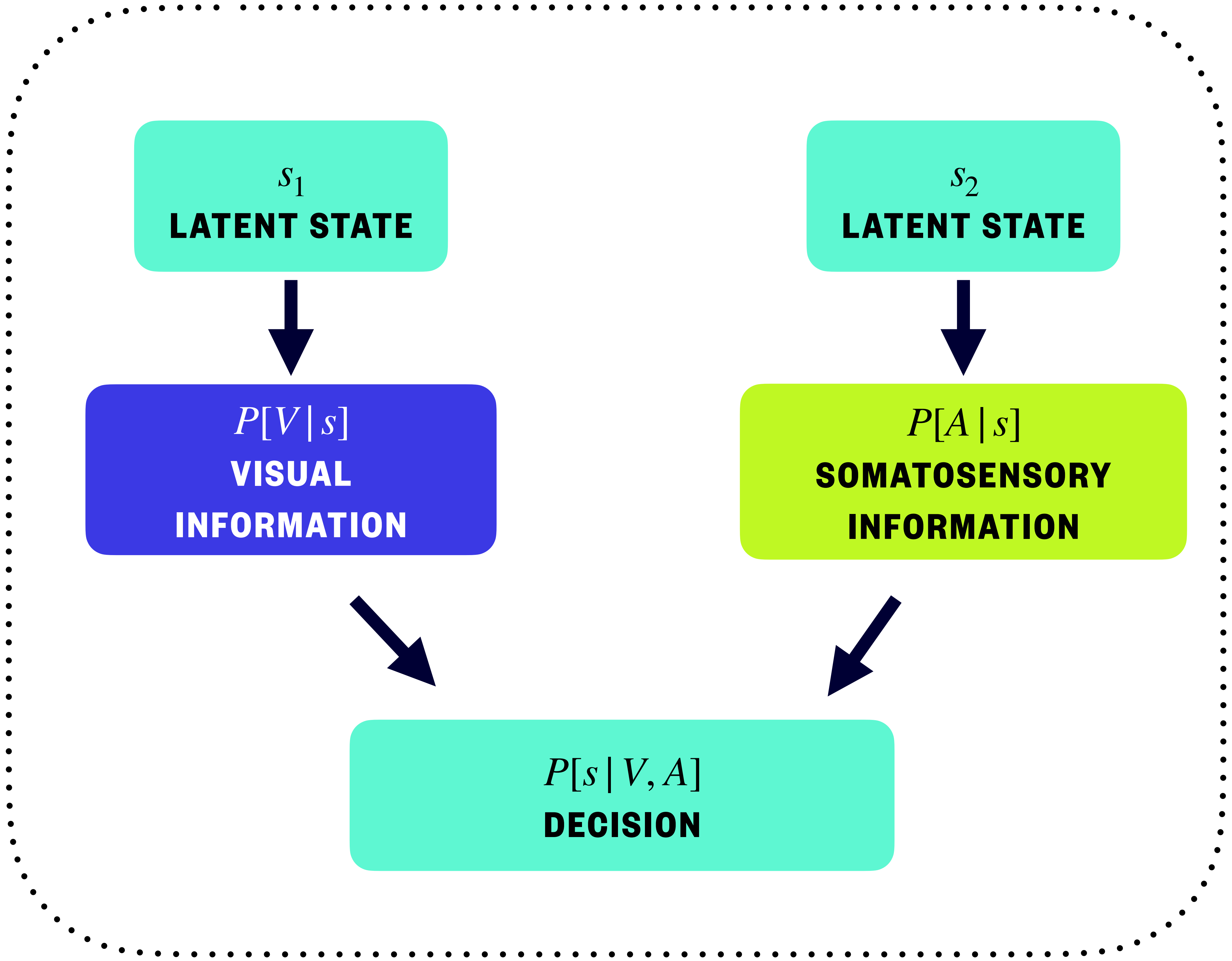
- **Causal Inference**

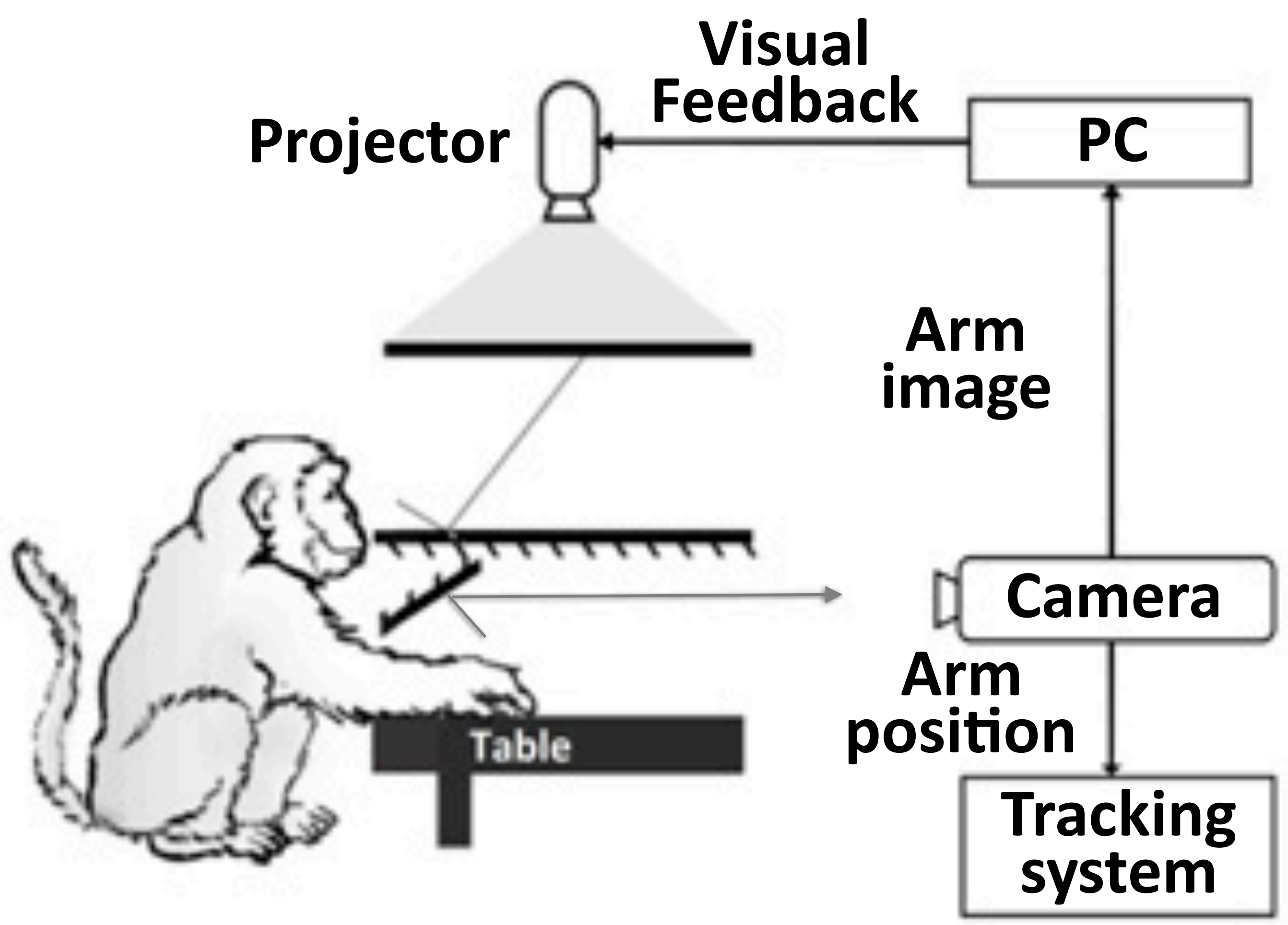
Rubber Hand Illusion

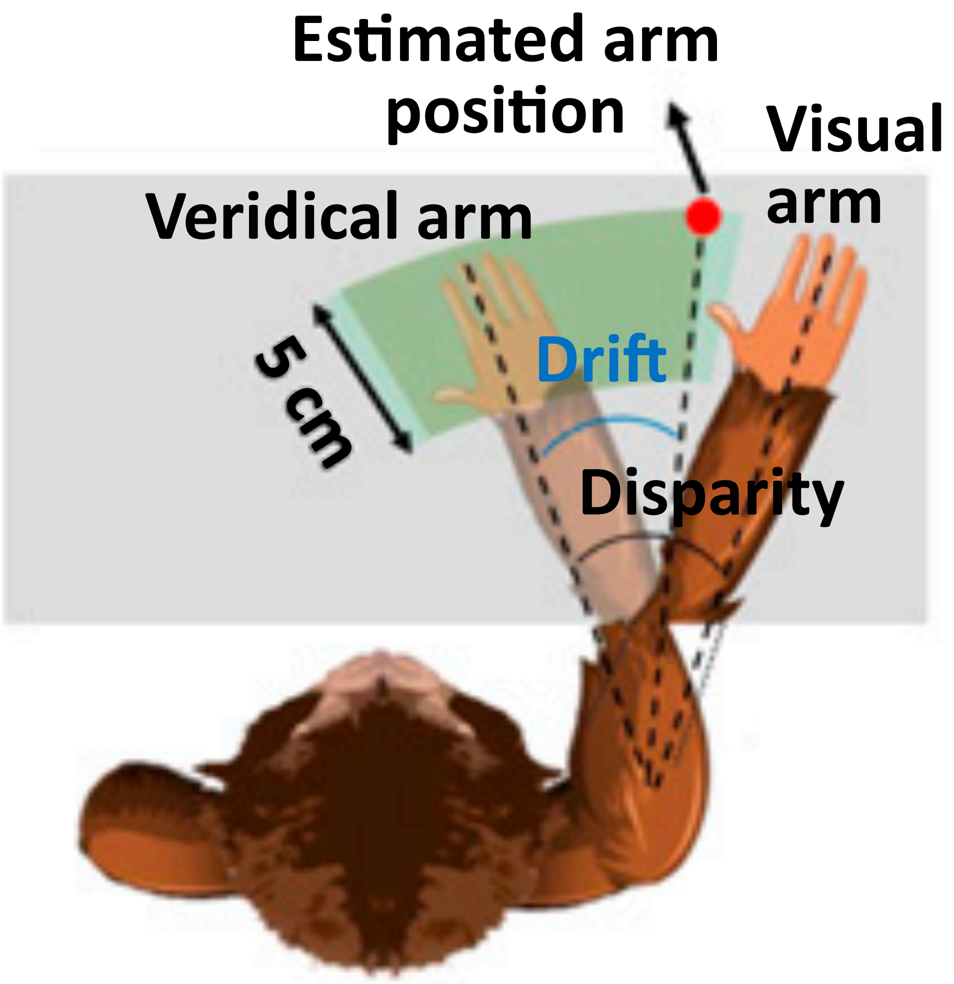
Causal Inference

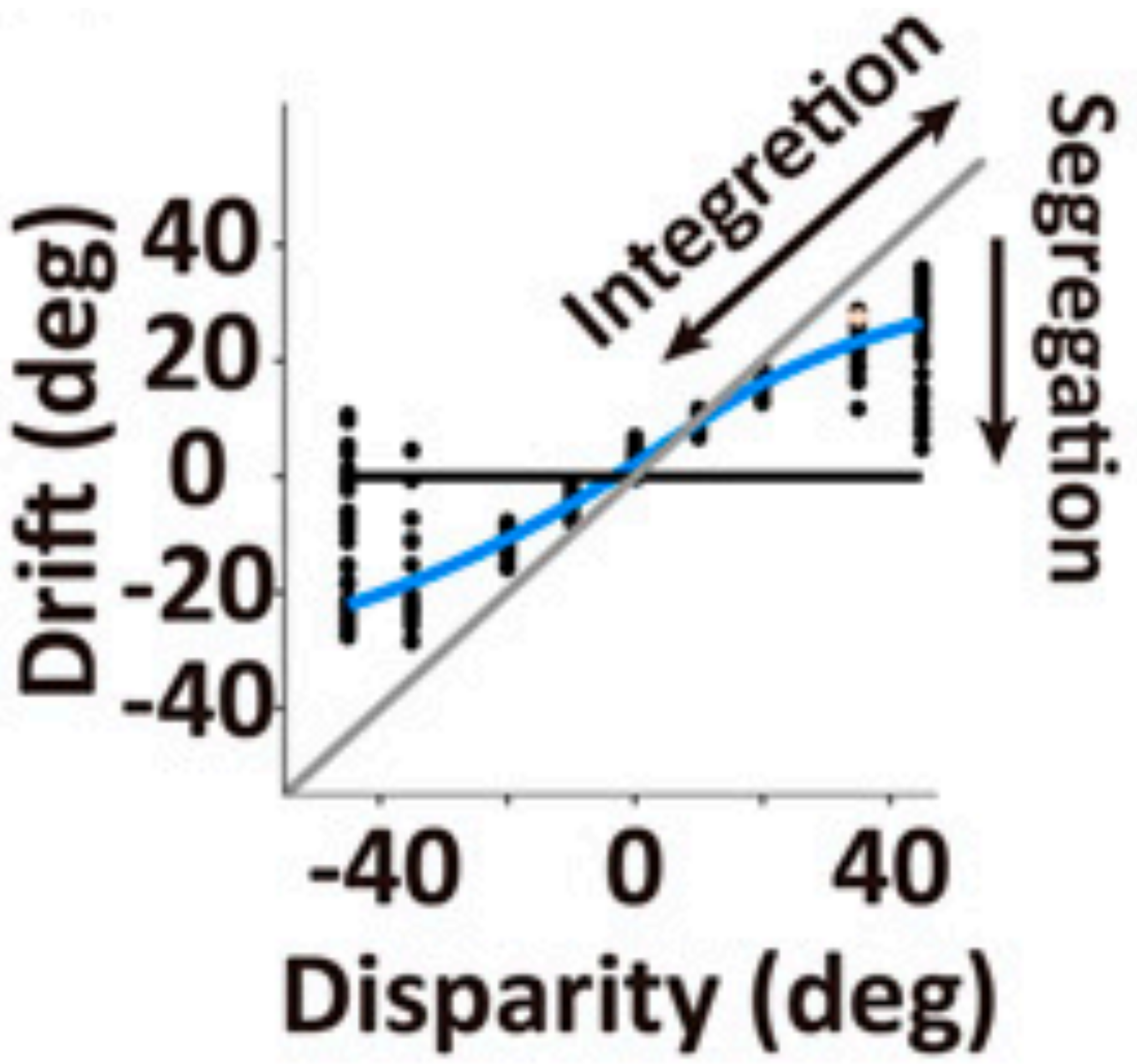




Experiment







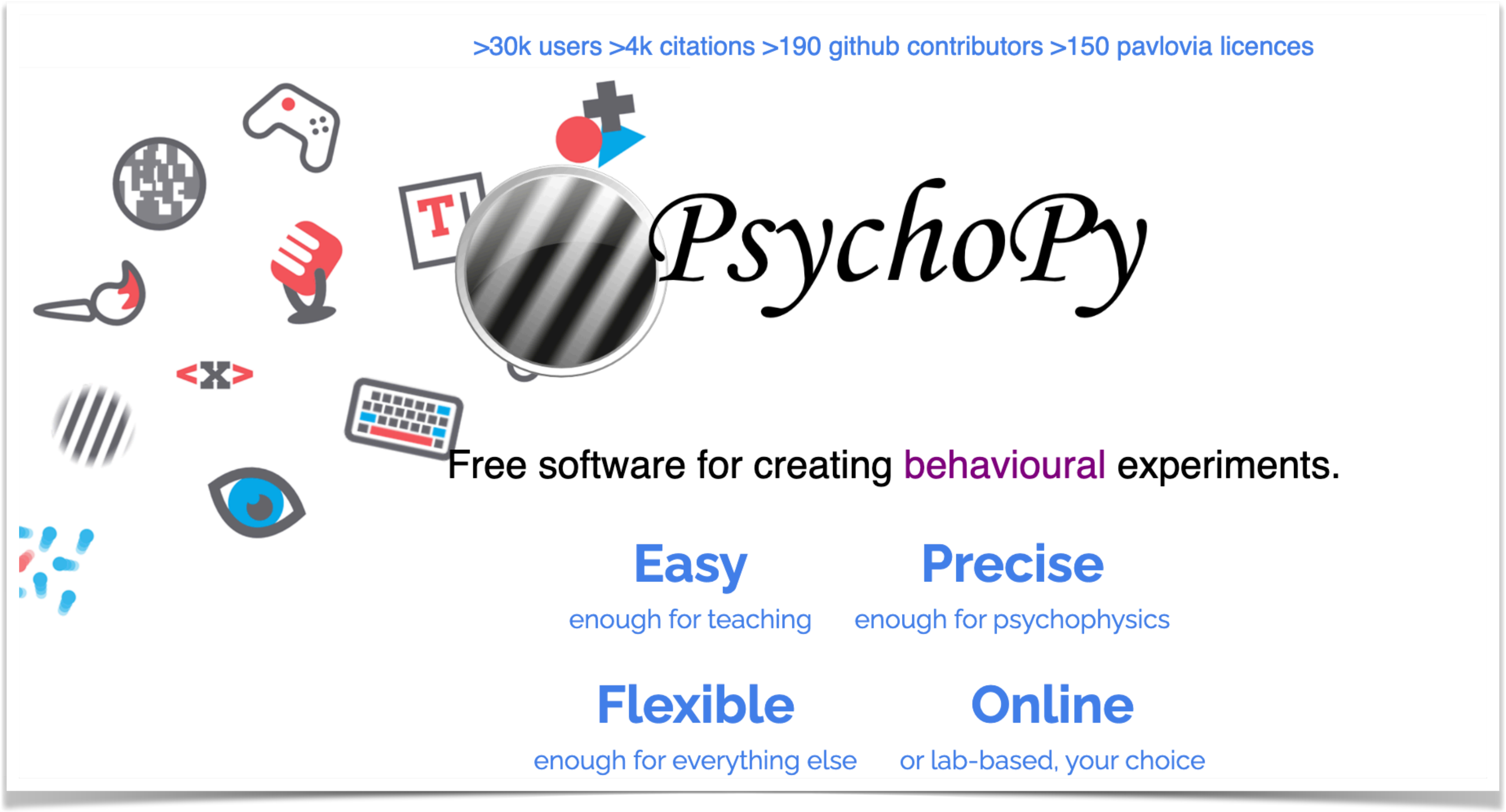
Fang et al., 2019

Summary

- Auditory
  - Hair cells, Frequency tuning, localization
- Vestibular
  - Hair cells, Mucula, Ampulla
- Somatosensory
  - Many subtypes, adaptation, pathways, homunculus
- Multimodal Integration
  - Bayesian integration
- Causal Inference
  - Bayesian, again

Homework

Learn PsychoPy, go through https://psychopy.org/coder/tutorial2.html, and write your own experiment.



Alternative Homework

Learn Psychtoolbox, go through https://peterscarfe.com/orientationThreshold.html, and write your own experiment.

Psychtoolbox-3

Psychophysics Toolbox Version 3 (PTB-3) is a free set of Matlab and GNU Octave functions for vision and neuroscience research. It makes it easy to synthesize and show accurately controlled visual and auditory stimuli and interact with the observer. Some of its functionality is available as part of Python toolkits like PsychoPy. For commercial support and services visit www.psychtoolbox.net. Follow us on Mastodon or Twitter @psychtoolbox